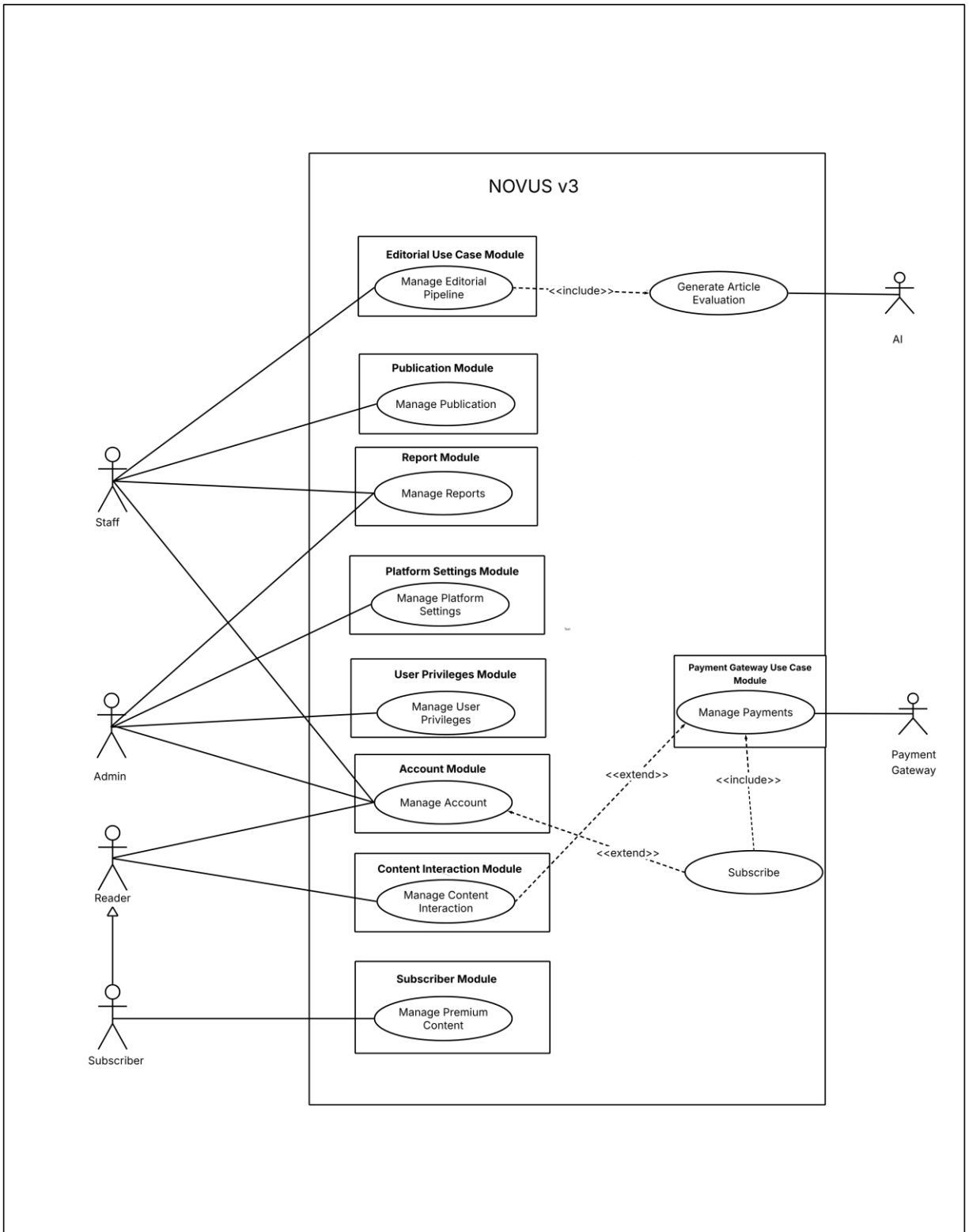


# **NOVUS**

## **FULLY DRESSED USE CASE SPECIFICATION**

BOSS Magazine PH  
IEEE-Aligned Black-and-White Format

**Figure 1. Overall NOVUS Use Case Diagram**



## 1. Document Overview

This document specifies the fully dressed use cases of the NOVUS web-based editorial and publishing management system for BOSS Magazine PH. Each use case is presented in an individual two-column table with the use case information, basic flow, alternate flows, and exception flows. The document uses a portrait-oriented, black-and-white IEEE-style presentation.

The current requirements baseline for Module 4, Manage Platform Settings, contains exactly three use cases: UC-4.1 Update Site Theme and Layout, UC-4.2 Configure System Integrations, and UC-4.3 Toggle Maintenance Mode. The previously removed Configure Global SEO Settings use case is not included.

UC-1.4 Request Revision supports both Writer-submitted articles and magazine designs uploaded by Graphics Designers. UC-1.8 is titled Upload Magazine Design and supports versioned design submissions and revision cycles. Module 9 contains payment-processing flows only and does not include promo-code or discount-code functionality.

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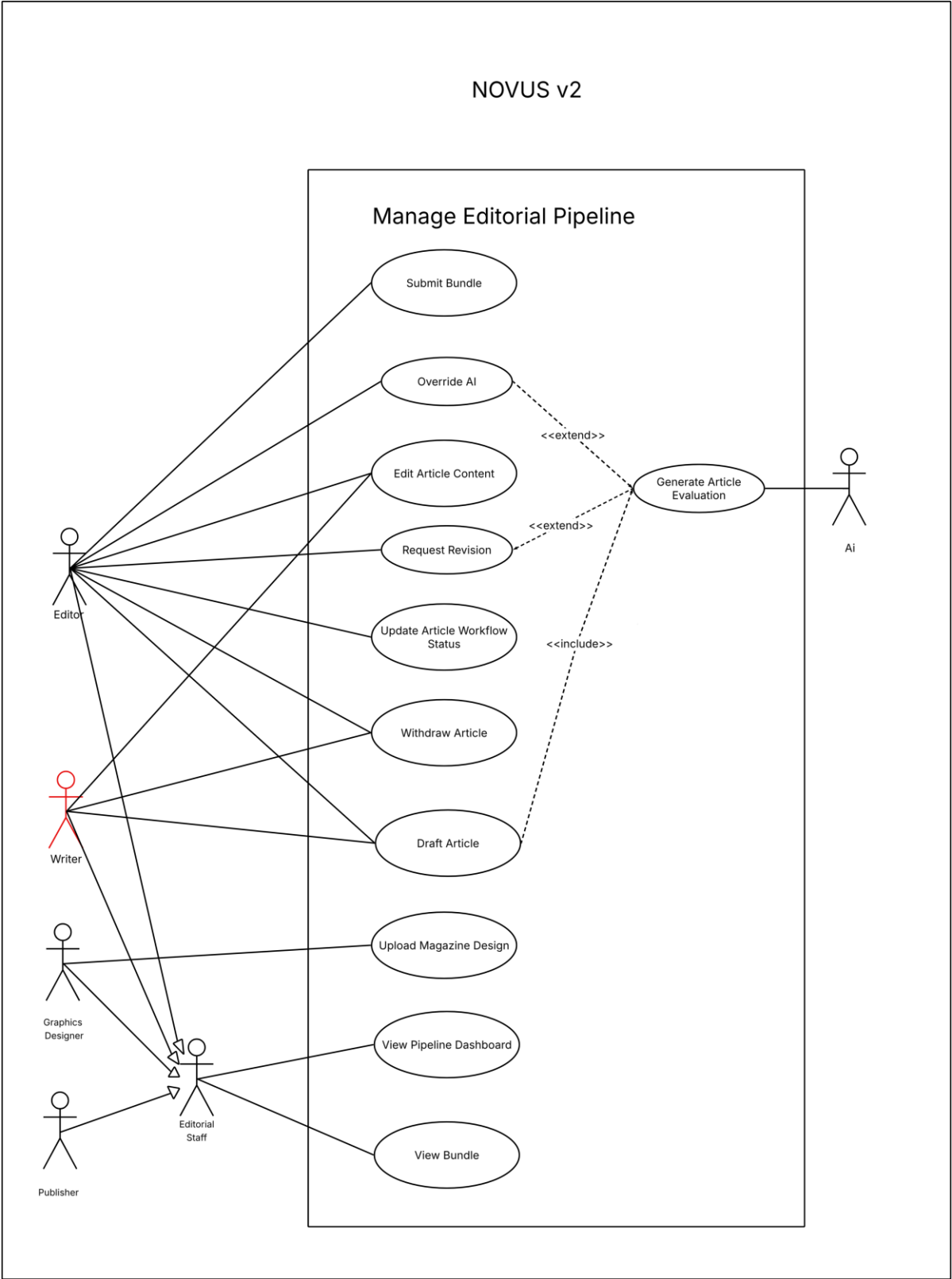
- UC-8.1 Download Digital Issues
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3.1 Module 1: Manage Editorial Pipeline

Figure 2. Manage Editorial Pipeline Use Case Diagram



UC-1.1: Submit Bundle

| USE CASE ELEMENT | SPECIFICATION |
|------------------|---------------|
|------------------|---------------|

| USE CASE ELEMENT                                               | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                                    | UC-1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Author                                                         | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Purpose                                                        | Allow an Editor to formally submit a completed content bundle (article draft, images, metadata) into the editorial pipeline for tracking and review.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Related Use Case                                               | Extends into: Update Article Workflow Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Requirement Traceability                                       | PR-EDT-01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Priority                                                       | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Preconditions                                                  | System is online<br>Editor is authenticated and logged in<br>A draft bundle exists and is marked ready by its Writer & Graphics Designer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Postconditions                                                 | Bundle status changes to "Submitted"<br>Bundle appears on the Pipeline Dashboard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Actors                                                         | Editor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| BASIC FLOW                                                     | <ol style="list-style-type: none"><li>1. Editor navigates to the Editorial Pipeline module.</li><li>2. Editor selects "Submit Bundle."</li><li>3. System displays draft bundles ready for submission.</li><li>4. Editor selects the target bundle and reviews its contents.</li><li>5. Editor clicks "Submit."</li><li>6. System validates that all required components (text, assets, metadata) are present.</li><li>7. System sets the bundle's workflow status to "Submitted" and timestamps the action.</li><li>8. System displays a confirmation notice to the Editor.</li></ol>                                                                                                                                         |
| ALTERNATE FLOW<br>A1 (Incomplete Bundle)                       | <ol style="list-style-type: none"><li>1. The user initiates the operation for the target record.</li><li>2. The system validates the required information and components.</li><li>3. The system detects that one or more required items are missing or incomplete.</li><li>4. The system identifies the missing items and displays corrective guidance.</li><li>5. The system prevents the incomplete operation from changing the workflow state.</li><li>6. The user supplies or corrects the missing information and retries.</li><li>7. The system revalidates the corrected record before continuing.</li></ol>                                                                                                           |
| EXCEPTION FLOW<br>E1 (Save Failure)                            | <ol style="list-style-type: none"><li>1. The user confirms the requested change.</li><li>2. The system validates the requested transition.</li><li>3. The system attempts to persist the change to the authoritative data store.</li><li>4. The persistence operation fails or times out.</li><li>5. The system retains the last confirmed state and does not report success.</li><li>6. The system logs the failure and displays a controlled retry message.</li><li>7. The user retries after service recovery.</li></ol>                                                                                                                                                                                                   |
| EXCEPTION FLOW<br>E-AUTH (Authorization<br>or Session Failure) | <ol style="list-style-type: none"><li>1. The user initiates the protected operation.</li><li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li><li>3. The system detects an expired session or missing permission.</li><li>4. The system blocks the state change and does not partially commit it.</li><li>5. The system records the denied attempt when required by security policy.</li><li>6. The system redirects to authentication or displays access denied.</li><li>7. After authorization is restored, the user may retry.</li></ol> <p>UC-1.2: Draft Article</p> <p>Use Case Name: Manage Editorial Pipeline_Draft Article</p> |

### UC-1.2: Draft Article

| USE CASE ELEMENT                                      | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                           | UC-1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Author                                                | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Purpose                                               | Allow a Writer & Editor to create and save a new article draft within the pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Related Use Case                                      | Includes: Generate Article Evaluation (UC-1.11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Requirement Traceability                              | PR-EDT-02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Priority                                              | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Preconditions                                         | System is online<br>Writer or Editor is authenticated and logged in<br>Writer has an assigned topic or pitch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Postconditions                                        | A new article draft exists with status "Drafting" or "Awaiting Evaluation"<br>An AI evaluation has been generated for the draft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Actors                                                | Writer<br>Editor<br>AI (secondary, via include)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| BASIC FLOW                                            | 1. Writer or Editor selects "Draft Article" from the Editorial Pipeline module.<br>2. System opens a blank article editor.<br>3. Writer or Editor enters the headline, body content, and relevant metadata (tags, category).<br>4. Writer or Editor saves the draft.<br>5. System stores the draft and assigns it a unique bundle reference.<br>6. System includes Generate Article Evaluation (UC-1.11) to automatically analyze the draft.<br>7. System displays the evaluation summary alongside the draft.                                                                                                                      |
| ALTERNATE FLOW<br>A1 (Save as Incomplete)             | 1. The user initiates the operation for the target record.<br>2. The system validates the required information and components.<br>3. The system detects that one or more required items are missing or incomplete.<br>4. The system identifies the missing items and displays corrective guidance.<br>5. The system prevents the incomplete operation from changing the workflow state.<br>6. The user supplies or corrects the missing information and retries.<br>7. The system revalidates the corrected record before continuing.                                                                                               |
| EXCEPTION FLOW<br>E1 (Evaluation Service Unavailable) | 1. The workflow reaches the point where an AI result is expected.<br>2. The system checks for a current result for the exact content version.<br>3. The system determines that the result is unavailable, incomplete, or delayed.<br>4. The system preserves the content and marks evaluation as pending/unavailable.<br>5. The system queues or retries the evaluation.<br>6. The system informs the Editor that manual review may be required.<br>7. When a valid result arrives, the system attaches it to the correct version.<br>UC-1.3: Edit Article Content<br>Use Case Name: Manage Editorial Pipeline_Edit Article Content |

### UC-1.3: Edit Article Content

| USE CASE ELEMENT | SPECIFICATION                                                                               |
|------------------|---------------------------------------------------------------------------------------------|
| Use Case ID      | UC-1.3                                                                                      |
| Author           | Novus Development Team                                                                      |
| Purpose          | Allow an Editor or Writer to modify the text content of an article already in the pipeline. |

| USE CASE ELEMENT                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Related Use Case                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Requirement Traceability                        | PR-EDT-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Priority                                        | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Preconditions                                   | System is online<br>User is authenticated as Editor or Writer<br>Target article exists in the pipeline and is not locked/published                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Postconditions                                  | Article content is updated<br>Edit history/version log is updated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Actors                                          | Editor<br>Writer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BASIC FLOW                                      | <ol style="list-style-type: none"> <li>1. User opens the target article bundle from the pipeline.</li> <li>2. User selects "Edit Content."</li> <li>3. System loads the article into an editable text editor.</li> <li>4. User modifies headline, body text, or metadata.</li> <li>5. User clicks "Save Changes."</li> <li>6. System validates the content and stores the new version.</li> <li>7. System logs the edit (author, timestamp) in the article's version history.</li> </ol>                                                                                                                                                            |
| ALTERNATE FLOW<br>A1 (Discard Changes)          | <ol style="list-style-type: none"> <li>1. The user chooses to cancel, discard, or leave the operation.</li> <li>2. The system checks for unsaved changes or a pending transaction.</li> <li>3. The system requests confirmation when the action could lose work or affect billing.</li> <li>4. The user confirms cancellation.</li> <li>5. The system discards only uncommitted changes or safely terminates the pending operation.</li> <li>6. The system restores the last confirmed state and returns the user to the appropriate screen.</li> <li>7. The system does not create an unintended version, state change, or entitlement.</li> </ol> |
| EXCEPTION FLOW<br>E1 (Concurrent Edit Conflict) | <ol style="list-style-type: none"> <li>1. The user opens the target and begins the operation.</li> <li>2. Another authorized session changes the same record.</li> <li>3. The system compares the held version with the authoritative version.</li> <li>4. The system detects stale data.</li> <li>5. The system prevents silent overwrite.</li> <li>6. The system displays the latest state or a comparison/merge option.</li> <li>7. The user reviews the current version and retries.</li> </ol> <p>UC-1.4: Request Revision</p> <p>Use Case Name: Manage Editorial Pipeline_Request Revision</p>                                                |

UC-1.4: Request Revision

| USE CASE ELEMENT         | SPECIFICATION                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-1.4                                                                                                                                                                |
| Author                   | Novus Development Team                                                                                                                                                |
| Purpose                  | Allow an Editor to request revisions for either a Writer-submitted article or a magazine design uploaded by a Graphics Designer, based on editorial or design review. |
| Related Use Case         | Extends: Generate Article Evaluation (UC-1.11) for articles; may also be initiated directly from the magazine-design review workflow.                                 |
| Requirement Traceability | PR-EDT-04                                                                                                                                                             |
| Priority                 | High                                                                                                                                                                  |
| Preconditions            | System is online                                                                                                                                                      |

| USE CASE ELEMENT                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Editor is authenticated and logged in</p> <p>Target article or magazine design exists and is in a reviewable state</p> <p>Editor has permission to request revision</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Postconditions                                  | <p>Target status changes to "Revision Requested"</p> <p>Current submitted version is preserved</p> <p>Revision notes are recorded</p> <p>Responsible Writer or Graphics Designer is notified</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Actors                                          | <p>Editor</p> <p>Writer (for article revision requests)</p> <p>Graphics Designer (for magazine-design revision requests)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| BASIC FLOW                                      | <ol style="list-style-type: none"> <li>1. Editor opens the submitted article or uploaded magazine design requiring review.</li> <li>2. System identifies the target type as either a Writer-submitted article or a Graphics Designer-uploaded magazine design.</li> <li>3. System displays the latest version, responsible contributor, current workflow status, and revision history.</li> <li>4. Editor selects "Request Revision."</li> <li>5. System prompts the Editor for specific revision notes appropriate to the selected target type.</li> <li>6. Editor enters actionable feedback. For articles, feedback may address content, structure, grammar, factual support, or editorial requirements. For magazine designs, feedback may address page layout, typography, imagery, branding, dimensions, or other design requirements.</li> <li>7. Editor reviews and confirms the revision request.</li> <li>8. System revalidates that the target remains reviewable and that the Editor still has permission.</li> <li>9. System preserves the submitted version and records the revision request, Editor, timestamp, target type, and feedback.</li> <li>10. System changes the target status to "Revision Requested."</li> <li>11. System notifies the responsible Writer for an article or Graphics Designer for a magazine design.</li> <li>12. System updates the pipeline/dashboard and makes the revision request visible to the responsible contributor.</li> </ol> |
| ALTERNATE FLOW<br>A1 (No AI Feedback Available) | <ol style="list-style-type: none"> <li>1. Editor opens an article for which the AI evaluation has not completed.</li> <li>2. System confirms that the article is still in a reviewable state.</li> <li>3. Editor performs manual editorial review without relying on an AI result.</li> <li>4. Editor selects "Request Revision" and enters specific revision notes.</li> <li>5. System records the request without requiring an AI evaluation reference.</li> <li>6. System changes the article status to "Revision Requested" and notifies the Writer.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| ALTERNATE FLOW<br>A2 (Magazine Design Revision) | <ol style="list-style-type: none"> <li>1. Editor opens a magazine design uploaded by a Graphics Designer.</li> <li>2. System displays the latest design version, design status, Designer, and prior revision history.</li> <li>3. Editor reviews the design against issue layout, branding, typography, imagery, dimensions, and other configured design requirements.</li> <li>4. Editor selects "Request Revision" and enters design-specific corrections.</li> <li>5. System validates that the design is still the current reviewable version.</li> <li>6. System records the revision request and changes the design status to "Revision Requested."</li> <li>7. System notifies the responsible Graphics Designer and exposes the revision notes in the design workspace.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| EXCEPTION FLOW<br>E1 (Notification Failure)     | <ol style="list-style-type: none"> <li>1. The system successfully records the revision request and changes the target status.</li> <li>2. The system creates a notification for the responsible Writer or Graphics Designer.</li> <li>3. The notification service fails to deliver the message.</li> <li>4. The system keeps the revision state and stored notes unchanged.</li> <li>5. The system records the failed delivery and marks the notification for retry or follow-up.</li> <li>6. The Editor can see the follow-up indicator on the dashboard.</li> <li>7. The system retries delivery when the notification service becomes available.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| EXCEPTION FLOW                                  | <ol style="list-style-type: none"> <li>1. Editor opens an article or magazine design and begins writing revision notes.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| USE CASE ELEMENT                       | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E2 (Target State Changed Concurrently) | <ol style="list-style-type: none"> <li>Another authorized user submits a newer version or changes the target status.</li> <li>Editor attempts to confirm the revision request.</li> <li>System detects that the target version or workflow state has changed.</li> <li>System prevents the Editor from attaching the revision request to a stale version.</li> <li>System displays the latest version and current status.</li> <li>Editor reviews the current version and submits a new revision request if it is still necessary.</li> </ol> UC-1.5: Override AI<br>Use Case Name: Manage Editorial Pipeline_Override AI |

UC-1.5: Override AI

| USE CASE ELEMENT                             | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                  | UC-1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Author                                       | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Purpose                                      | Allow an Editor to manually override the AI's evaluation verdict for an article, using editorial judgment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Related Use Case                             | Extends: Generate Article Evaluation (UC-1.11) at extension point "OverrideRequested"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Requirement Traceability                     | PR-EDT-05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Priority                                     | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Preconditions                                | System is online<br>Editor is authenticated and logged in<br>Generate Article Evaluation has produced a verdict the Editor disagrees with                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Postconditions                               | Article's evaluation verdict is flagged as "Editor Override"<br>Override justification is recorded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Actors                                       | Editor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BASIC FLOW                                   | <ol style="list-style-type: none"> <li>Editor reviews the AI evaluation verdict for an article.</li> <li>Editor selects "Override AI."</li> <li>System requires the Editor to enter a justification for the override.</li> <li>Editor submits the justification.</li> <li>System updates the article's evaluation record to reflect the manual override.</li> <li>System allows the article to proceed in the pipeline despite the original AI verdict.</li> </ol>                                                                                                                                                                                                                                                                        |
| ALTERNATE FLOW<br>A1 (Partial Override)      | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                   |
| EXCEPTION FLOW<br>E1 (Missing Justification) | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol> UC-1.6: Update Article Workflow Status<br>Use Case Name: Manage Editorial Pipeline_Update Article Workflow Status |

## UC-1.6: Update Article Workflow Status

| USE CASE ELEMENT                                            | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                                 | UC-1.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Author                                                      | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Purpose                                                     | Allow an Editor to move an article through its editorial workflow stages (e.g., In Review, Approved, Ready for Publication).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Related Use Case                                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Requirement Traceability                                    | PR-EDT-06                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Priority                                                    | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Preconditions                                               | System is online<br>Editor is authenticated and logged in<br>Article exists in the pipeline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Postconditions                                              | Article's workflow status is updated<br>Status change is logged in the article's history                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Actors                                                      | Editor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| BASIC FLOW                                                  | <ol style="list-style-type: none"> <li>1. Editor opens the target article's pipeline entry.</li> <li>2. Editor selects "Update Workflow Status."</li> <li>3. System displays available next-stage options based on the current status.</li> <li>4. Editor selects the new status.</li> <li>5. Editor confirms the change.</li> <li>6. System updates the article's status and records the transition with a timestamp.</li> </ol>                                                                                                                                                                                                                                                                                                           |
| ALTERNATE FLOW<br>A1 (Revert Status)                        | <ol style="list-style-type: none"> <li>1. The AI or workflow produces a result qualifying for the alternate approval path.</li> <li>2. The system verifies the result belongs to the current version.</li> <li>3. The system checks the configured threshold and mandatory-review rules.</li> <li>4. If all conditions pass, the system advances the workflow.</li> <li>5. The system records the rule-based advancement.</li> <li>6. The system notifies the next responsible role.</li> <li>7. If human review remains mandatory, the system leaves the item in review.</li> </ol>                                                                                                                                                        |
| EXCEPTION FLOW<br>E1 (Invalid Transition)                   | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                |
| EXCEPTION FLOW<br>E-AUTH (Authorization or Session Failure) | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-1.7: Withdraw Article</p> <p>Use Case Name: Manage Editorial Pipeline_Withdraw Article</p> |

## UC-1.7: Withdraw Article

| USE CASE ELEMENT                                            | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                                 | UC-1.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Author                                                      | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Purpose                                                     | Allow an Editor or Writer to remove an article from the active pipeline before it is published.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Related Use Case                                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Requirement Traceability                                    | PR-EDT-07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Priority                                                    | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Preconditions                                               | System is online<br>User is authenticated as Editor or Writer<br>Article is not yet published                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Postconditions                                              | Article status changes to "Withdrawn"<br>Article is removed from active pipeline views                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Actors                                                      | Editor<br>Writer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| BASIC FLOW                                                  | <ol style="list-style-type: none"> <li>1. User opens the target article's pipeline entry.</li> <li>2. User selects "Withdraw Article."</li> <li>3. System prompts for confirmation and a withdrawal reason.</li> <li>4. User confirms.</li> <li>5. System sets the article status to "Withdrawn" and logs the reason.</li> <li>6. System removes the article from active pipeline dashboards.</li> </ol>                                                                                                                                                                                                                                                                                                                                                 |
| ALTERNATE FLOW<br>A1 (Restore Withdrawn Article)            | <ol style="list-style-type: none"> <li>1. The user selects the withdrawal operation.</li> <li>2. The system retrieves the latest authoritative state.</li> <li>3. The system checks whether withdrawal is still permitted.</li> <li>4. If permitted, the system requests confirmation and any required reason.</li> <li>5. The user confirms.</li> <li>6. The system records the withdrawal and updates the state.</li> <li>7. If publication is already irreversible, the system blocks withdrawal and directs the user to the appropriate post-publication process.</li> </ol>                                                                                                                                                                         |
| EXCEPTION FLOW<br>E1 (Already Published)                    | <ol style="list-style-type: none"> <li>1. The user selects the withdrawal operation.</li> <li>2. The system retrieves the latest authoritative state.</li> <li>3. The system checks whether withdrawal is still permitted.</li> <li>4. If permitted, the system requests confirmation and any required reason.</li> <li>5. The user confirms.</li> <li>6. The system records the withdrawal and updates the state.</li> <li>7. If publication is already irreversible, the system blocks withdrawal and directs the user to the appropriate post-publication process.</li> </ol>                                                                                                                                                                         |
| EXCEPTION FLOW<br>E-AUTH (Authorization or Session Failure) | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-1.8: Upload Magazine Design</p> <p>Use Case Name: Manage Editorial Pipeline_ Upload Magazine Design</p> |

## UC-1.8: Upload Magazine Design

| USE CASE ELEMENT                                                 | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                                      | UC-1.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Author                                                           | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Purpose                                                          | Allow a Graphics Designer to upload a magazine design for an issue, create a versioned submission, and provide an Editor with a design that can be reviewed and returned for revision.                                                                                                                                                                                                                                                                                                                                                                                              |
| Related Use Case                                                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Requirement Traceability                                         | PR-EDT-08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Priority                                                         | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Preconditions                                                    | System is online<br>Graphics Designer is authenticated and logged in<br>Target issue exists<br>Designer is assigned or authorized to upload to the target issue<br>Design file meets configured publication requirements                                                                                                                                                                                                                                                                                                                                                            |
| Postconditions                                                   | Design file is securely stored as a versioned asset<br>Design is linked to the target issue<br>New version is submitted for Editor review<br>Prior versions remain traceable<br>Editor is notified                                                                                                                                                                                                                                                                                                                                                                                  |
| Actors                                                           | Graphics Designer<br>Editor (secondary reviewer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BASIC FLOW                                                       | User navigates to the specific issue or bundle record in the system.<br>User selects "Upload Magazine Designs."<br>System opens a secure file upload interface.<br>User selects the compiled digital magazine file (e.g., PDF) from their local device.<br>System validates the file format and ensures it is within the maximum file size limits.<br>System securely uploads the file, attaches it to the issue record, and timestamps the action.<br>System displays a success confirmation notice to the user.                                                                   |
| ALTERNATE FLOW<br>A1 (Version Replacement)                       | 1. Graphics Designer opens an issue that already has a previously uploaded design.<br>2. System displays the active design version and prior version history.<br>3. Graphics Designer selects a corrected design file.<br>4. System validates Designer permission, target issue, file type, file size, and integrity.<br>5. System stores the corrected file as a new version rather than silently destroying the previous version.<br>6. System marks the new version as the active design submission.<br>7. System notifies the Editor that a revised design is ready for review. |
| ALTERNATE FLOW<br>A2 (Upload After Revision Request)             | 1. Graphics Designer opens a design with status "Revision Requested."<br>2. System displays the Editor's latest revision notes and the version being revised.<br>3. Graphics Designer prepares a corrected design.<br>4. Graphics Designer uploads the corrected file.<br>5. System creates a new version linked to the same revision cycle.<br>6. System preserves the prior submitted version and its revision notes.<br>7. System changes the new version to "Submitted for Review" and notifies the Editor.                                                                     |
| EXCEPTION FLOW<br>E1 (Unsupported Format or Size Limit Exceeded) | 1. Graphics Designer selects a design file.<br>2. System validates the file extension, detected content type, and file size before storage.<br>3. System detects an unsupported format or a file exceeding the configured size limit.<br>4. System rejects the file and does not create a new design version.<br>5. System displays the accepted formats and maximum file size.<br>6. Graphics Designer exports or selects a compliant file.<br>7. Graphics Designer retries the upload.                                                                                            |

| USE CASE ELEMENT                                        | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXCEPTION FLOW E2 (Upload Interruption)</b>          | <ol style="list-style-type: none"> <li>1. Graphics Designer begins transferring a valid design file.</li> <li>2. The network connection is interrupted during the transfer.</li> <li>3. System detects that the file was not completely transferred.</li> <li>4. System marks the upload as failed or resumable according to storage capability.</li> <li>5. System prevents the incomplete file from becoming the active design version.</li> <li>6. System informs the Graphics Designer that the upload must be retried or resumed.</li> <li>7. Graphics Designer retries after connectivity is restored.</li> </ol>                                                                                                                                                                                                                                                                                |
| <b>EXCEPTION FLOW E3 (Unauthorized Target Issue)</b>    | <ol style="list-style-type: none"> <li>1. Graphics Designer selects an issue for upload.</li> <li>2. System verifies that the Designer is assigned or authorized for the selected issue.</li> <li>3. System determines that the Designer does not have permission for the target issue.</li> <li>4. System blocks the upload before storing the file.</li> <li>5. System displays an access-denied message without exposing restricted issue details.</li> <li>6. Graphics Designer selects an authorized issue or requests assignment from the appropriate staff member.</li> </ol>                                                                                                                                                                                                                                                                                                                   |
| <b>EXCEPTION FLOW E4 (Concurrent Design Submission)</b> | <ol style="list-style-type: none"> <li>1. Graphics Designer opens the issue while another authorized Designer is preparing a newer design version.</li> <li>2. The other Designer submits a new version before the first upload is finalized.</li> <li>3. System detects the changed active version when the first Designer confirms the upload.</li> <li>4. System does not silently overwrite the newer version.</li> <li>5. System creates the first Designer's file as a separate version only if versioning rules permit it; otherwise it asks the Designer to refresh.</li> <li>6. System displays the current active version and latest submission history.</li> <li>7. Graphics Designer reviews the current state and retries using the latest version information.</li> </ol> <p>UC-1.9: View Pipeline Dashboard</p> <p>Use Case Name: Manage Editorial Pipeline_View Pipeline Dashboard</p> |

### UC-1.9: View Pipeline Dashboard

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-1.9                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Author</b>                   | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Purpose</b>                  | Allow Editorial Staff to view an overview of all bundles currently active in the editorial pipeline, including status and assignees.                                                                                                                                                                                                                                                                                                |
| <b>Related Use Case</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Requirement Traceability</b> | PR-EDT-09                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Priority</b>                 | Medium                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Preconditions</b>            | System is online<br>User is authenticated as Editor, Writer, Publisher or Graphics Designer                                                                                                                                                                                                                                                                                                                                         |
| <b>Postconditions</b>           | Dashboard is displayed reflecting current pipeline state                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Actors</b>                   | Editorial Staff (Editor, Writer, Graphics Designer, Publisher)                                                                                                                                                                                                                                                                                                                                                                      |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. Editorial Staff member navigates to the Editorial Pipeline module.</li> <li>2. Editorial Staff selects "View Pipeline Dashboard."</li> <li>3. System retrieves all active bundles and their current statuses.</li> <li>4. System displays the dashboard with filters (by status, assignee, deadline).</li> <li>5. Editorial Staff member may apply filters to narrow the view.</li> </ol> |
| <b>ALTERNATE FLOW</b>           | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> </ol>                                                                                                                                                                                                                                                                                                                         |

| USE CASE ELEMENT                                 | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A1 (No Active Bundles)</b>                    | <ol style="list-style-type: none"> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>   |
| <b>EXCEPTION FLOW<br/>E1 (Data Load Failure)</b> | <ol style="list-style-type: none"> <li>1. The user opens the dashboard.</li> <li>2. The system verifies session and permissions.</li> <li>3. The system queries current records.</li> <li>4. If no records match, the system displays the configured empty state.</li> <li>5. If retrieval fails, the system displays a controlled error and retry option.</li> <li>6. The system logs the failure when appropriate.</li> <li>7. The user retries or changes the filter.</li> </ol> <p>UC-1.10: View Bundle</p> <p>Use Case Name: Manage Editorial Pipeline_View Bundle</p> |

### UC-1.10: View Bundle

| USE CASE ELEMENT                                          | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                                        | UC-1.10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Author</b>                                             | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose</b>                                            | Allow Editorial Staff to view the full detail of a single content bundle, including its content, assets, status, and history.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Related Use Case</b>                                   | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Requirement Traceability</b>                           | PR-EDT-10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Priority</b>                                           | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Preconditions</b>                                      | <p>System is online</p> <p>User is authenticated as Editor, Writer, Graphics Designer, or Publisher</p> <p>Target bundle exists</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Postconditions</b>                                     | Bundle detail view is displayed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Actors</b>                                             | Editorial Staff (Editor, Writer, Graphics Designer, Publisher)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>BASIC FLOW</b>                                         | <ol style="list-style-type: none"> <li>1. Editorial Staff member selects a bundle from the Pipeline Dashboard or search.</li> <li>2. System retrieves the bundle's content, assets, status, and revision history.</li> <li>3. System displays the full bundle detail view.</li> <li>4. Editorial Staff member reviews the information.</li> </ol>                                                                                                                                                                                                                                                                                                                                                    |
| <b>ALTERNATE FLOW<br/>A1 (Navigate to Related Action)</b> | <ol style="list-style-type: none"> <li>1. The user selects a file and begins upload.</li> <li>2. The system validates role, target assignment, file type, file size, and upload rules.</li> <li>3. The system rejects invalid files before finalization.</li> <li>4. For a valid file, the system transfers it to controlled storage.</li> <li>5. The system verifies transfer completion and file integrity.</li> <li>6. The system creates a new version and preserves prior versions.</li> <li>7. If interrupted, the system prevents the incomplete file from becoming active and marks it failed or resumable.</li> <li>8. The user retries or resumes after correcting the problem.</li> </ol> |
| <b>EXCEPTION FLOW<br/>E1 (Bundle Not Found)</b>           | <ol style="list-style-type: none"> <li>1. The user opens the dashboard.</li> <li>2. The system verifies session and permissions.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| USE CASE ELEMENT | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | 3. The system queries current records.<br>4. If no records match, the system displays the configured empty state.<br>5. If retrieval fails, the system displays a controlled error and retry option.<br>6. The system logs the failure when appropriate.<br>7. The user retries or changes the filter.<br>UC-1.11: Generate Article Evaluation<br>Use Case Name: Manage Editorial Pipeline_Generate Article Evaluation |

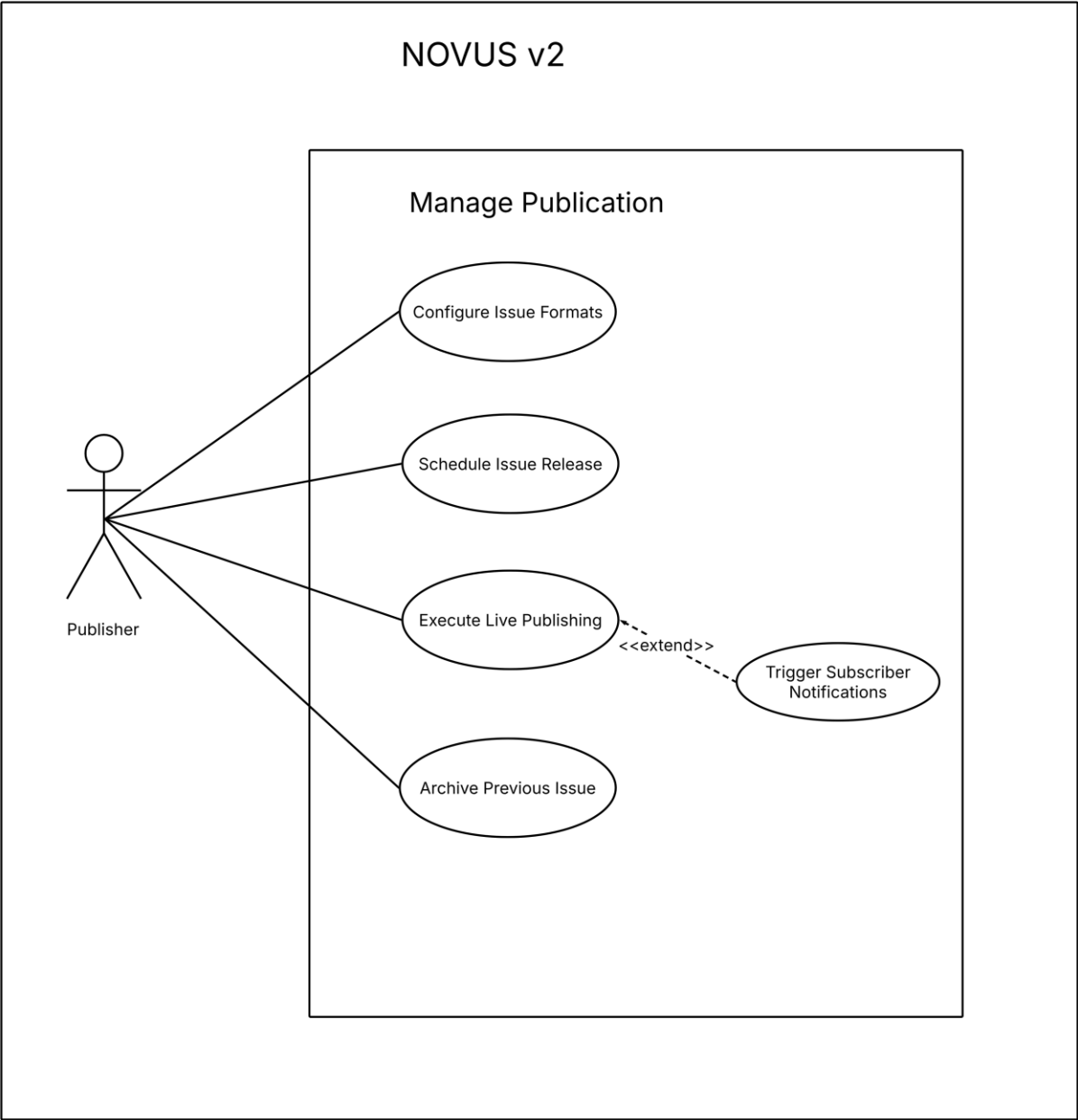
UC-1.11: Generate Article Evaluation

| USE CASE ELEMENT                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                     | UC-1.11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Author                                          | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Purpose                                         | Allow an Editor to request revisions for either a Writer-submitted article or a magazine design uploaded by a Graphics Designer, based on editorial or design review.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Related Use Case                                | Extends: Generate Article Evaluation (UC-1.11) for articles; may also be initiated directly from the magazine-design review workflow.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Requirement Traceability                        | PR-EDT-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Priority                                        | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Preconditions                                   | System is online<br>Editor is authenticated and logged in<br>Target article or magazine design exists and is in a reviewable state<br>Editor has permission to request revision                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Postconditions                                  | Target status changes to "Revision Requested"<br>Current submitted version is preserved<br>Revision notes are recorded<br>Responsible Writer or Graphics Designer is notified                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Actors                                          | Editor<br>Writer (for article revision requests)<br>Graphics Designer (for magazine-design revision requests)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| BASIC FLOW                                      | 1. System sends the article's content to the AI evaluation service (triggered via include from Draft Article).<br>2. AI analyzes the article for grammar, and readability.<br>3. AI returns a quality score and a recommendation (Approve / Reject).<br>4. System attaches the evaluation result to the article record.<br>5. System exposes extension point "RevisionRequested" to the Editor if the recommendation is "Reject."<br>6. System exposes extension point "OverrideRequested" to the Editor for any evaluation result.<br>7. System notifies the Editor that evaluation is complete. |
| ALTERNATE FLOW<br>A1 (High-Confidence Approval) | 1. The AI or workflow produces a result qualifying for the alternate approval path.<br>2. The system verifies the result belongs to the current version.<br>3. The system checks the configured threshold and mandatory-review rules.<br>4. If all conditions pass, the system advances the workflow.<br>5. The system records the rule-based advancement.<br>6. The system notifies the next responsible role.<br>7. If human review remains mandatory, the system leaves the item in review.                                                                                                    |
| EXCEPTION FLOW<br>E1 (AI Service Timeout)       | 1. The workflow reaches the point where an AI result is expected.<br>2. The system checks for a current result for the exact content version.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| USE CASE ELEMENT | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>3. The system determines that the result is unavailable, incomplete, or delayed.</p> <p>4. The system preserves the content and marks evaluation as pending/unavailable.</p> <p>5. The system queues or retries the evaluation.</p> <p>6. The system informs the Editor that manual review may be required.</p> <p>7. When a valid result arrives, the system attaches it to the correct version.</p> <p>MODULE 2: Manage Publication</p> <p>Actors: Publisher</p> <p>UC-2.1: Configure Issue Formats</p> <p>Use Case Name: Manage Publication_Configure Issue Formats</p> |

### 3.2 Module 2: Manage Publication

Figure 3. Manage Publication Use Case Diagram



### UC-2.1: Configure Issue Formats

| USE CASE ELEMENT                          | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                               | UC-2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Author                                    | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Purpose                                   | Allow the Publisher to define and configure the output formats (digital layout templates, PDF/e-reader formats) available for a publication issue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Related Use Case                          | <ul style="list-style-type: none"><li>None</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Requirement Traceability                  | PR-PUB-01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Priority                                  | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Preconditions                             | <ul style="list-style-type: none"><li>System is online</li><li>Publisher is authenticated and logged in</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Postconditions                            | <ul style="list-style-type: none"><li>Issue format configuration is saved and available for scheduling</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Actors                                    | <ul style="list-style-type: none"><li>Publisher</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BASIC FLOW                                | <ol style="list-style-type: none"><li>Publisher navigates to the Manage Publication module.</li><li>Publisher selects “Configure Issue Formats.”</li><li>System displays existing format templates (e.g., Web, PDF, ePub).</li><li>Publisher creates a new format or edits an existing one (dimensions, cover template, file type).</li><li>Publisher saves the configuration.</li><li>System validates and stores the format settings.</li></ol>                                                                                                                                                                                                                                                                                                                         |
| ALTERNATE FLOW<br>A1 (Duplicate Format)   | <ol style="list-style-type: none"><li>The user selects a file and begins upload.</li><li>The system validates role, target assignment, file type, file size, and upload rules.</li><li>The system rejects invalid files before finalization.</li><li>For a valid file, the system transfers it to controlled storage.</li><li>The system verifies transfer completion and file integrity.</li><li>The system creates a new version and preserves prior versions.</li><li>If interrupted, the system prevents the incomplete file from becoming active and marks it failed or resumable.</li><li>The user retries or resumes after correcting the problem.</li></ol>                                                                                                       |
| EXCEPTION FLOW<br>E1 (Invalid Parameters) | <ol style="list-style-type: none"><li>The user selects a file and begins upload.</li><li>The system validates role, target assignment, file type, file size, and upload rules.</li><li>The system rejects invalid files before finalization.</li><li>For a valid file, the system transfers it to controlled storage.</li><li>The system verifies transfer completion and file integrity.</li><li>The system creates a new version and preserves prior versions.</li><li>If interrupted, the system prevents the incomplete file from becoming active and marks it failed or resumable.</li><li>The user retries or resumes after correcting the problem.</li></ol> <p>UC-2.2: Schedule Issue Release</p> <p>Use Case Name: Manage Publication_Schedule Issue Release</p> |

### UC-2.2: Schedule Issue Release

| USE CASE ELEMENT | SPECIFICATION                                                                    |
|------------------|----------------------------------------------------------------------------------|
| Use Case ID      | UC-2.2                                                                           |
| Author           | Novus Development Team                                                           |
| Purpose          | Allow the Publisher to set a future release date and time for an upcoming issue. |

| USE CASE ELEMENT                                            | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Related Use Case                                            | <ul style="list-style-type: none"> <li>None</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Requirement Traceability                                    | PR-PUB-02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Priority                                                    | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Preconditions                                               | <ul style="list-style-type: none"> <li>System is online</li> <li>Issue content (approved bundles) is ready</li> <li>Issue format is configured</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Postconditions                                              | <ul style="list-style-type: none"> <li>Issue is queued with a scheduled release date/time</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Actors                                                      | <ul style="list-style-type: none"> <li>Publisher</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| BASIC FLOW                                                  | <ol style="list-style-type: none"> <li>Publisher selects an issue that is ready for release.</li> <li>Publisher selects “Schedule Issue Release.”</li> <li>System displays a calendar/time picker.</li> <li>Publisher selects the release date and time.</li> <li>Publisher confirms.</li> <li>System sets the issue status to “Scheduled” and stores the release timestamp.</li> </ol>                                                                                                                                                                                                                                                                                                                                       |
| ALTERNATE FLOW<br>A1 (Reschedule)                           | <ol style="list-style-type: none"> <li>The user enters a date/time.</li> <li>The system validates format and scheduling rules.</li> <li>The system detects that the value violates a constraint.</li> <li>The system rejects the value and explains the rule.</li> <li>The user enters a compliant future value.</li> <li>The system revalidates and continues.</li> </ol>                                                                                                                                                                                                                                                                                                                                                    |
| EXCEPTION FLOW<br>E1 (Invalid Date)                         | <ol style="list-style-type: none"> <li>The user enters a date/time.</li> <li>The system validates format and scheduling rules.</li> <li>The system detects that the value violates a constraint.</li> <li>The system rejects the value and explains the rule.</li> <li>The user enters a compliant future value.</li> <li>The system revalidates and continues.</li> </ol>                                                                                                                                                                                                                                                                                                                                                    |
| EXCEPTION FLOW<br>E-AUTH (Authorization or Session Failure) | <ol style="list-style-type: none"> <li>The user initiates the protected operation.</li> <li>The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>The system detects an expired session or missing permission.</li> <li>The system blocks the state change and does not partially commit it.</li> <li>The system records the denied attempt when required by security policy.</li> <li>The system redirects to authentication or displays access denied.</li> <li>After authorization is restored, the user may retry.</li> </ol> <p>UC-2.3: Execute Live Publishing</p> <p>Use Case Name: Manage Publication_Execute Live Publishing</p> |

UC-2.3: Execute Live Publishing

| USE CASE ELEMENT         | SPECIFICATION                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-2.3                                                                                                                |
| Author                   | Novus Development Team                                                                                                |
| Purpose                  | Allow the Publisher to manually trigger immediate publishing of an issue, bypassing or overriding a schedule.         |
| Related Use Case         | <ul style="list-style-type: none"> <li>Extended by: Trigger Subscriber Notifications (UC-2.3.1) — optional</li> </ul> |
| Requirement Traceability | PR-PUB-03                                                                                                             |

| USE CASE ELEMENT                                         | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority                                                 | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Preconditions                                            | <ul style="list-style-type: none"> <li>System is online</li> <li>Issue is fully approved and ready</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Postconditions                                           | <ul style="list-style-type: none"> <li>Issue is published live and accessible per its access tier</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Actors                                                   | <ul style="list-style-type: none"> <li>Publisher</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| BASIC FLOW                                               | <ol style="list-style-type: none"> <li>Publisher opens a ready issue.</li> <li>Publisher selects “Execute Live Publishing.”</li> <li>System prompts for confirmation.</li> <li>Publisher confirms.</li> <li>System publishes the issue immediately, updates status to “Published,” and makes it live on the platform. At this point, the optional Trigger Subscriber Notifications extension (UC-2.3.1) may fire.</li> <li>System logs the publishing event.</li> </ol>                                                                                                                                                                                                                                                                           |
| ALTERNATE FLOW A (Flow)                                  | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                           |
| EXCEPTION FLOW E1 (Publishing Failure)                   | <ol style="list-style-type: none"> <li>The user initiates the operation for the target record.</li> <li>The system validates the required information and components.</li> <li>The system detects that one or more required items are missing or incomplete.</li> <li>The system identifies the missing items and displays corrective guidance.</li> <li>The system prevents the incomplete operation from changing the workflow state.</li> <li>The user supplies or corrects the missing information and retries.</li> <li>The system revalidates the corrected record before continuing.</li> </ol>                                                                                                                                            |
| EXCEPTION FLOW E-AUTH (Authorization or Session Failure) | <ol style="list-style-type: none"> <li>The user initiates the protected operation.</li> <li>The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>The system detects an expired session or missing permission.</li> <li>The system blocks the state change and does not partially commit it.</li> <li>The system records the denied attempt when required by security policy.</li> <li>The system redirects to authentication or displays access denied.</li> <li>After authorization is restored, the user may retry.</li> </ol> <p>UC-2.3.1: Trigger Subscriber Notifications</p> <p>Use Case Name: Manage Publication_Trigger Subscriber Notifications</p> |

### UC-2.3.1: Trigger Subscriber Notifications

| USE CASE ELEMENT         | SPECIFICATION                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-2.3.1                                                                                                                                                    |
| Author                   | Novus Development Team                                                                                                                                      |
| Purpose                  | Allow the Publisher to send notifications to subscribers and readers about a new issue or article, as an optional extension of the live publishing process. |
| Related Use Case         | <ul style="list-style-type: none"> <li>Extends: Execute Live Publishing (UC-2.3)</li> </ul>                                                                 |
| Requirement Traceability | PR-PUB-04                                                                                                                                                   |

| USE CASE ELEMENT                             | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority                                     | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Preconditions                                | <ul style="list-style-type: none"> <li>System is online</li> <li>Execute Live Publishing (UC-2.3) has reached the point where the issue is published or is about to be published</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Postconditions                               | <ul style="list-style-type: none"> <li>Notification is dispatched to the relevant subscriber segment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Actors                                       | <ul style="list-style-type: none"> <li>Publisher</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| BASIC FLOW                                   | <ol style="list-style-type: none"> <li>Immediately after Execute Live Publishing (UC-2.3) publishes the issue, System offers the Publisher the option to trigger subscriber notifications.</li> <li>Publisher elects to proceed with the extension.</li> <li>System displays notification channel options (email, push, in-app).</li> <li>Publisher customizes the notification message and target segment.</li> <li>Publisher confirms send.</li> <li>System dispatches the notifications, logs delivery statistics, and returns control to Execute Live Publishing (UC-2.3).</li> </ol>                                                                                 |
| ALTERNATE FLOW<br>A1 (Schedule Notification) | <ol style="list-style-type: none"> <li>The user enters a date/time.</li> <li>The system validates format and scheduling rules.</li> <li>The system detects that the value violates a constraint.</li> <li>The system rejects the value and explains the rule.</li> <li>The user enters a compliant future value.</li> <li>The system revalidates and continues.</li> </ol>                                                                                                                                                                                                                                                                                                |
| EXCEPTION FLOW<br>E1 (Delivery Failure)      | <ol style="list-style-type: none"> <li>The user initiates the operation for the target record.</li> <li>The system validates the required information and components.</li> <li>The system detects that one or more required items are missing or incomplete.</li> <li>The system identifies the missing items and displays corrective guidance.</li> <li>The system prevents the incomplete operation from changing the workflow state.</li> <li>The user supplies or corrects the missing information and retries.</li> <li>The system revalidates the corrected record before continuing.</li> </ol>                                                                    |
| EXCEPTION FLOW<br>E2 (Extension Declined)    | <ol style="list-style-type: none"> <li>The user confirms publication.</li> <li>The system validates publish readiness.</li> <li>The system starts the publication transaction and records the pre-publication state.</li> <li>The publishing service returns an error, timeout, or incomplete result.</li> <li>The system verifies what, if anything, reached the live environment.</li> <li>The system does not report full publication when verification is incomplete.</li> <li>The system records the failure and exposes controlled recovery/retry.</li> </ol> <p>UC-2.5: Archive Previous Issue</p> <p>Use Case Name: Manage Publication_Archive Previous Issue</p> |

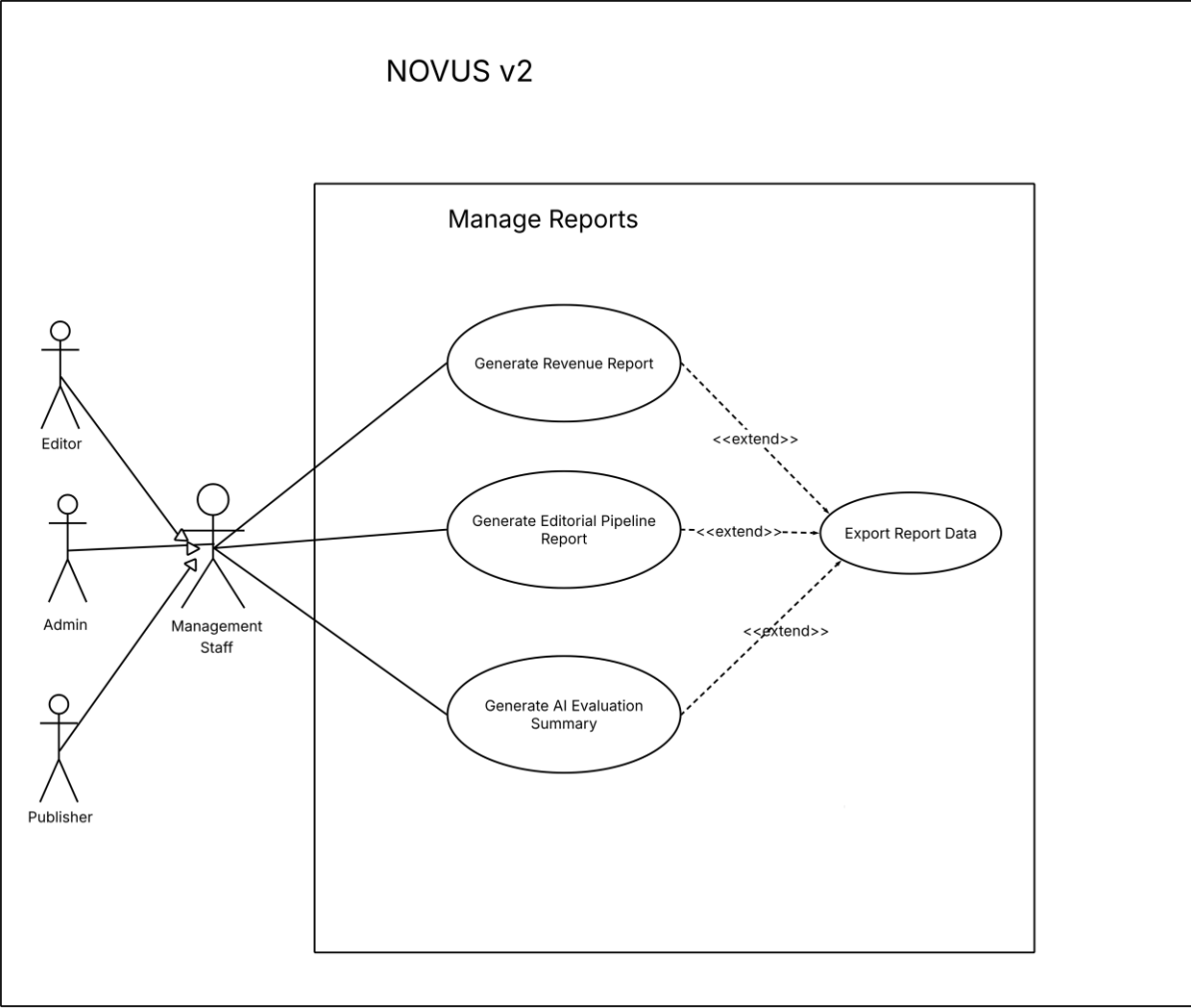
### UC-2.5: Archive Previous Issue

| USE CASE ELEMENT         | SPECIFICATION                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-2.5                                                                                                |
| Author                   | Novus Development Team                                                                                |
| Purpose                  | Allow the Publisher to move an older, already-published issue into the archive for historical access. |
| Related Use Case         | <ul style="list-style-type: none"> <li>None</li> </ul>                                                |
| Requirement Traceability | PR-PUB-05                                                                                             |
| Priority                 | Low                                                                                                   |

| USE CASE ELEMENT                                            | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preconditions                                               | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Issue has been published and is no longer the current issue</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Postconditions                                              | <ul style="list-style-type: none"> <li>• Issue is flagged as archived and relocated to the archive listing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Actors                                                      | <ul style="list-style-type: none"> <li>• Publisher</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| BASIC FLOW                                                  | <ol style="list-style-type: none"> <li>1. Publisher navigates to the issue list.</li> <li>2. Publisher selects an old issue.</li> <li>3. Publisher selects “Archive Previous Issue.”</li> <li>4. System prompts for confirmation.</li> <li>5. Publisher confirms.</li> <li>6. System updates the issue status to “Archived” and relocates it to the archive section.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| ALTERNATE FLOW<br>A1 (Bulk Archive)                         | <ol style="list-style-type: none"> <li>1. The user confirms publication.</li> <li>2. The system validates publish readiness.</li> <li>3. The system starts the publication transaction and records the pre-publication state.</li> <li>4. The publishing service returns an error, timeout, or incomplete result.</li> <li>5. The system verifies what, if anything, reached the live environment.</li> <li>6. The system does not report full publication when verification is incomplete.</li> <li>7. The system records the failure and exposes controlled recovery/retry.</li> </ol>                                                                                                                                                                                                                                                                                                                   |
| EXCEPTION FLOW<br>E1 (Active Issue Protection)              | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                                                                                                                                                               |
| EXCEPTION FLOW<br>E-AUTH (Authorization or Session Failure) | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>MODULE 3: Manage Reports</p> <p>Actors: Editor; Admin; Publisher; Management Staff (generalized parent of Editor, Admin, and Publisher)</p> <p>UC-3.1: Generate Revenue Report</p> <p>Use Case Name: Manage Reports_Generate Subscription Revenue Report</p> |

3.3 Module 3: Manage Reports

Figure 4. Manage Reports Use Case Diagram



**UC-3.1: Generate Revenue Report**

| USE CASE ELEMENT         | SPECIFICATION                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-3.1                                                                                                                                                                                                                                                                                          |
| Author                   | Novus Development Team                                                                                                                                                                                                                                                                          |
| Purpose                  | Allow Management Staff to generate a report summarizing subscription and payment revenue over a selected period.                                                                                                                                                                                |
| Related Use Case         | <ul style="list-style-type: none"><li>Extends into: Export Report Data (UC-3.1.1)</li></ul>                                                                                                                                                                                                     |
| Requirement Traceability | PR-RPT-01                                                                                                                                                                                                                                                                                       |
| Priority                 | High                                                                                                                                                                                                                                                                                            |
| Preconditions            | <ul style="list-style-type: none"><li>System is online</li><li>User is authenticated as Editor, Admin, or Publisher</li><li>Payment data exists for the requested period</li></ul>                                                                                                              |
| Postconditions           | <ul style="list-style-type: none"><li>Report is generated and available for viewing/export</li></ul>                                                                                                                                                                                            |
| Actors                   | <ul style="list-style-type: none"><li>Management Staff (Editor, Admin, Publisher)</li></ul>                                                                                                                                                                                                     |
| BASIC FLOW               | <ol style="list-style-type: none"><li>Management Staff navigates to the Manage Reports module.</li><li>Management Staff selects “Generate Subscription Revenue Report.”</li><li>System prompts for a date range and filters (plan type, region).</li><li>User submits the parameters.</li></ol> |

| USE CASE ELEMENT                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | 5. System compiles revenue data from payment records.<br>6. System displays the report on screen; Management Staff may optionally trigger the Export Report Data extension (UC-3.1.1) to download the report as PDF or CSV.                                                                                                                                                                                               |
| <b>ALTERNATE FLOW<br/>A1 (Recurring Report)</b> | 1. The user enters a date/time.<br>2. The system validates format and scheduling rules.<br>3. The system detects that the value violates a constraint.<br>4. The system rejects the value and explains the rule.<br>5. The user enters a compliant future value.<br>6. The system revalidates and continues.                                                                                                              |
| <b>EXCEPTION FLOW<br/>E1 (No Data Found)</b>    | 1. The user requests the selected list/report.<br>2. The system queries the current data set.<br>3. The system finds no matching records.<br>4. The system displays the empty state or configured fallback.<br>5. The system does not present fabricated or stale data.<br>6. The user adjusts filters or returns to another section.<br>UC-3.1.1: Export Report Data<br>Use Case Name: Manage Reports_Export Report Data |

### UC-3.1.1: Export Report Data

| USE CASE ELEMENT                                       | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                                     | UC-3.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Author</b>                                          | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose</b>                                         | Serve as a shared extension that converts the currently displayed report's on-screen data into a downloadable PDF or CSV file.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Related Use Case</b>                                | <ul style="list-style-type: none"> <li>• Extends: Generate Subscription Revenue Report (UC-3.1), Generate Editorial Pipeline Report (UC-3.2), Generate AI Evaluation Summary (UC-3.3)</li> </ul>                                                                                                                                                                                                                                                                                                                              |
| <b>Requirement Traceability</b>                        | PR-RPT-04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Priority</b>                                        | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Preconditions</b>                                   | <ul style="list-style-type: none"> <li>• System is online</li> <li>• A report has been generated and is currently displayed by UC-3.1, UC-3.2, or UC-3.3</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |
| <b>Postconditions</b>                                  | <ul style="list-style-type: none"> <li>• Report data is exported and downloaded in the selected format (PDF or CSV)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Actors</b>                                          | <ul style="list-style-type: none"> <li>• Management Staff (Editor, Admin, Publisher)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>BASIC FLOW</b>                                      | 1. While viewing a generated report, Management Staff selects "Export."<br>2. System displays the available export formats (PDF, CSV).<br>3. Management Staff selects a format.<br>4. System converts the on-screen report data into the selected file format.<br>5. System generates the downloadable file and returns control to the originating report use case (UC-3.1, UC-3.2, or UC-3.3).                                                                                                                               |
| <b>ALTERNATE FLOW<br/>A1 (Default Format Download)</b> | 1. The user selects a file and begins upload.<br>2. The system validates role, target assignment, file type, file size, and upload rules.<br>3. The system rejects invalid files before finalization.<br>4. For a valid file, the system transfers it to controlled storage.<br>5. The system verifies transfer completion and file integrity.<br>6. The system creates a new version and preserves prior versions.<br>7. If interrupted, the system prevents the incomplete file from becoming active and marks it failed or |

| USE CASE ELEMENT                              | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXCEPTION FLOW<br/>E1 (Export Failure)</b> | resumable.<br>8. The user retries or resumes after correcting the problem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                               | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>UC-3.2: Generate Editorial Pipeline Report</p> <p>Use Case Name: Manage Reports_Generate Editorial Pipeline Report</p> |

### UC-3.2: Generate Editorial Pipeline Report

| USE CASE ELEMENT                                  | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                                | UC-3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Author</b>                                     | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose</b>                                    | Allow Management Staff to generate a report on editorial pipeline throughput, such as articles drafted, submitted, published, and average turnaround time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Related Use Case</b>                           | <ul style="list-style-type: none"> <li>• Extends into: Export Report Data (UC-3.1.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Requirement Traceability</b>                   | PR-RPT-02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Priority</b>                                   | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Preconditions</b>                              | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Pipeline activity data exists</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Postconditions</b>                             | <ul style="list-style-type: none"> <li>• Report is generated and available for viewing/export</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Actors</b>                                     | <ul style="list-style-type: none"> <li>• Management Staff (Editor, Admin, Publisher)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>BASIC FLOW</b>                                 | <ol style="list-style-type: none"> <li>1. Management Staff selects “Generate Editorial Pipeline Report.”</li> <li>2. System prompts for a date range and scope (team or author).</li> <li>3. User submits the parameters.</li> <li>4. System compiles pipeline metrics.</li> <li>5. System displays the report on screen; Management Staff may optionally trigger the Export Report Data extension (UC-3.1.1) to download the report.</li> </ol>                                                                                                                                                                                             |
| <b>ALTERNATE FLOW<br/>A1 (Drill-Down)</b>         | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |
| <b>EXCEPTION FLOW<br/>E1 (Generation Timeout)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |

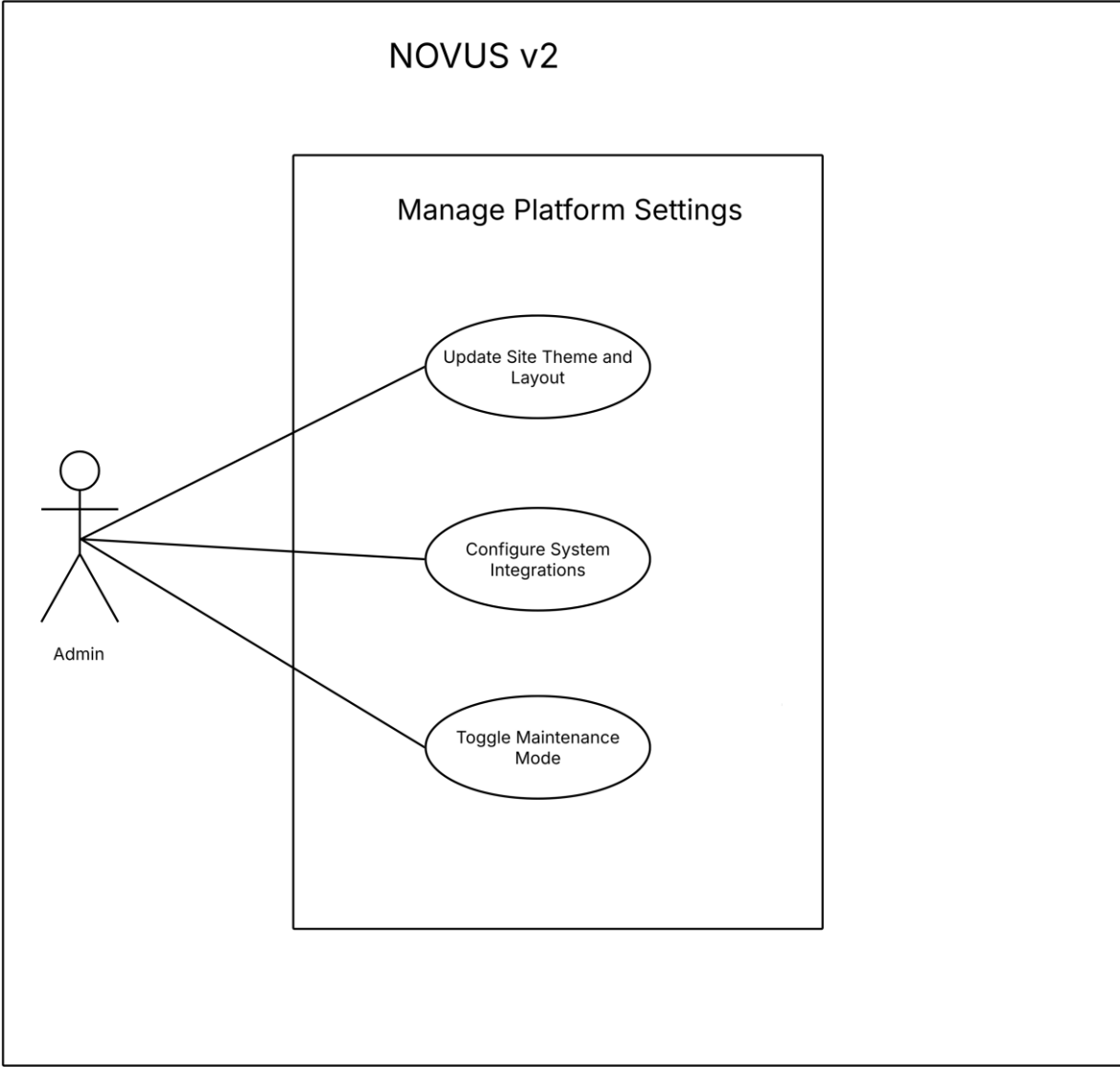
| USE CASE ELEMENT | SPECIFICATION                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------|
|                  | UC-3.3: Generate AI Evaluation Summary<br>Use Case Name: Manage Reports_Generate AI Evaluation Summary |

UC-3.3: Generate AI Evaluation Summary

| USE CASE ELEMENT                         | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                              | UC-3.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Author                                   | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Purpose                                  | Allow Management Staff to generate a summary report of AI article evaluations, including scores, override rate, and revision-request rate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Related Use Case                         | <ul style="list-style-type: none"> <li>Extends into: Export Report Data (UC-3.1.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Requirement Traceability                 | PR-RPT-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Priority                                 | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Preconditions                            | <ul style="list-style-type: none"> <li>System is online</li> <li>AI evaluations exist for the requested period</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Postconditions                           | <ul style="list-style-type: none"> <li>Report is generated and available for viewing/export</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Actors                                   | <ul style="list-style-type: none"> <li>Management Staff (Editor, Admin, Publisher)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BASIC FLOW                               | 1. Management Staff selects “Generate AI Evaluation Summary.”<br>2. System prompts for a date range.<br>3. User submits the parameters.<br>4. System aggregates evaluation scores, override counts, and revision-request counts.<br>5. System displays the summary report on screen; Management Staff may optionally trigger the Export Report Data extension (UC-3.1.1) to download the report.                                                                                                                                                                                                                                                                                                                              |
| ALTERNATE FLOW<br>A1 (Filter by Staff)   | 1. The user reaches the condition described by this flow.<br>2. The system detects the condition during the relevant basic-flow step.<br>3. The system validates the current record, permission, and business rules.<br>4. The system performs the permitted alternate handling or blocks the invalid operation.<br>5. The system explains the resulting state and required next action.<br>6. The system records the event when audit/security rules require it.<br>7. The user continues from the appropriate point after the condition is resolved.                                                                                                                                                                        |
| EXCEPTION FLOW<br>E1 (Insufficient Data) | 1. The user reaches the condition described by this flow.<br>2. The system detects the condition during the relevant basic-flow step.<br>3. The system validates the current record, permission, and business rules.<br>4. The system performs the permitted alternate handling or blocks the invalid operation.<br>5. The system explains the resulting state and required next action.<br>6. The system records the event when audit/security rules require it.<br>7. The user continues from the appropriate point after the condition is resolved.<br>MODULE 4: Manage Platform Settings<br>Actors: Admin<br>UC-4.1: Update Site Theme and Layout<br>Use Case Name: Manage Platform Settings_Update Site Theme and Layout |

3.4 Module 4: Manage Platform Settings

Figure 5. Manage Platform Settings Use Case Diagram



**UC-4.1: Update Site Theme and Layout**

| USE CASE ELEMENT         | SPECIFICATION                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-4.1                                                                                                         |
| Author                   | Novus Development Team                                                                                         |
| Purpose                  | Allow the Admin to customize the platform's visual theme and homepage layout.                                  |
| Related Use Case         | None                                                                                                           |
| Requirement Traceability | PR-SET-01                                                                                                      |
| Priority                 | Low                                                                                                            |
| Preconditions            | System is online<br>Admin is authenticated and logged in                                                       |
| Postconditions           | Theme and layout changes are applied site-wide                                                                 |
| Actors                   | Admin                                                                                                          |
| BASIC FLOW               | 1. Admin navigates to the Manage Platform Settings module.<br>2. Admin selects "Update Site Theme and Layout." |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <ol style="list-style-type: none"> <li>3. System displays current theme and layout options.</li> <li>4. Admin adjusts colors, and section ordering.</li> <li>5. Admin previews the changes.</li> <li>6. Admin saves.</li> <li>7. System applies the changes site-wide.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>ALTERNATE FLOW<br/>A1 (Revert to Default)</b>                    | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                          |
| <b>EXCEPTION FLOW<br/>E1 (Invalid Configuration)</b>                | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                          |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-4.2: Configure System Integrations</p> <p>Use Case Name: Manage Platform Settings_ Configure System Integrations</p> |

### UC-4.2: Configure System Integrations

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-4.2                                                                                                                                                                                                                                  |
| <b>Author</b>                   | Novus Development Team                                                                                                                                                                                                                  |
| <b>Purpose</b>                  | Allow the Admin to manage third-party integrations such as the payment gateway, and the email service.                                                                                                                                  |
| <b>Related Use Case</b>         | None                                                                                                                                                                                                                                    |
| <b>Requirement Traceability</b> | PR-SET-02                                                                                                                                                                                                                               |
| <b>Priority</b>                 | High                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | System is online<br>Admin is authenticated and logged in                                                                                                                                                                                |
| <b>Postconditions</b>           | Integration configuration is updated and connection verified                                                                                                                                                                            |
| <b>Actors</b>                   | Admin                                                                                                                                                                                                                                   |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. Admin selects "Configure System Integrations."</li> <li>2. System lists available integrations and their connection status.</li> <li>3. Admin selects an integration to configure.</li> </ol> |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <ol style="list-style-type: none"> <li>Admin enters or updates the API credentials or settings.</li> <li>Admin tests the connection.</li> <li>Admin saves.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>ALTERNATE FLOW<br/>A1 (Disable Integration)</b>                  | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                             |
| <b>EXCEPTION FLOW<br/>E1 (Connection Test Failure)</b>              | <ol style="list-style-type: none"> <li>The user submits credentials.</li> <li>The system validates them against the account.</li> <li>The system determines the credentials are incorrect.</li> <li>The system rejects the operation without revealing sensitive details.</li> <li>The system applies rate limiting or lockout rules when thresholds are reached.</li> <li>The system displays the permitted recovery option.</li> <li>The user retries or starts recovery.</li> </ol>                                                                                                                                                                                                                                              |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>The user initiates the protected operation.</li> <li>The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>The system detects an expired session or missing permission.</li> <li>The system blocks the state change and does not partially commit it.</li> <li>The system records the denied attempt when required by security policy.</li> <li>The system redirects to authentication or displays access denied.</li> <li>After authorization is restored, the user may retry.</li> </ol> <p>UC-4.3: Toggle Maintenance Mode</p> <p>Use Case Name: Manage Platform Settings_Toggle Maintenance Mode</p> |

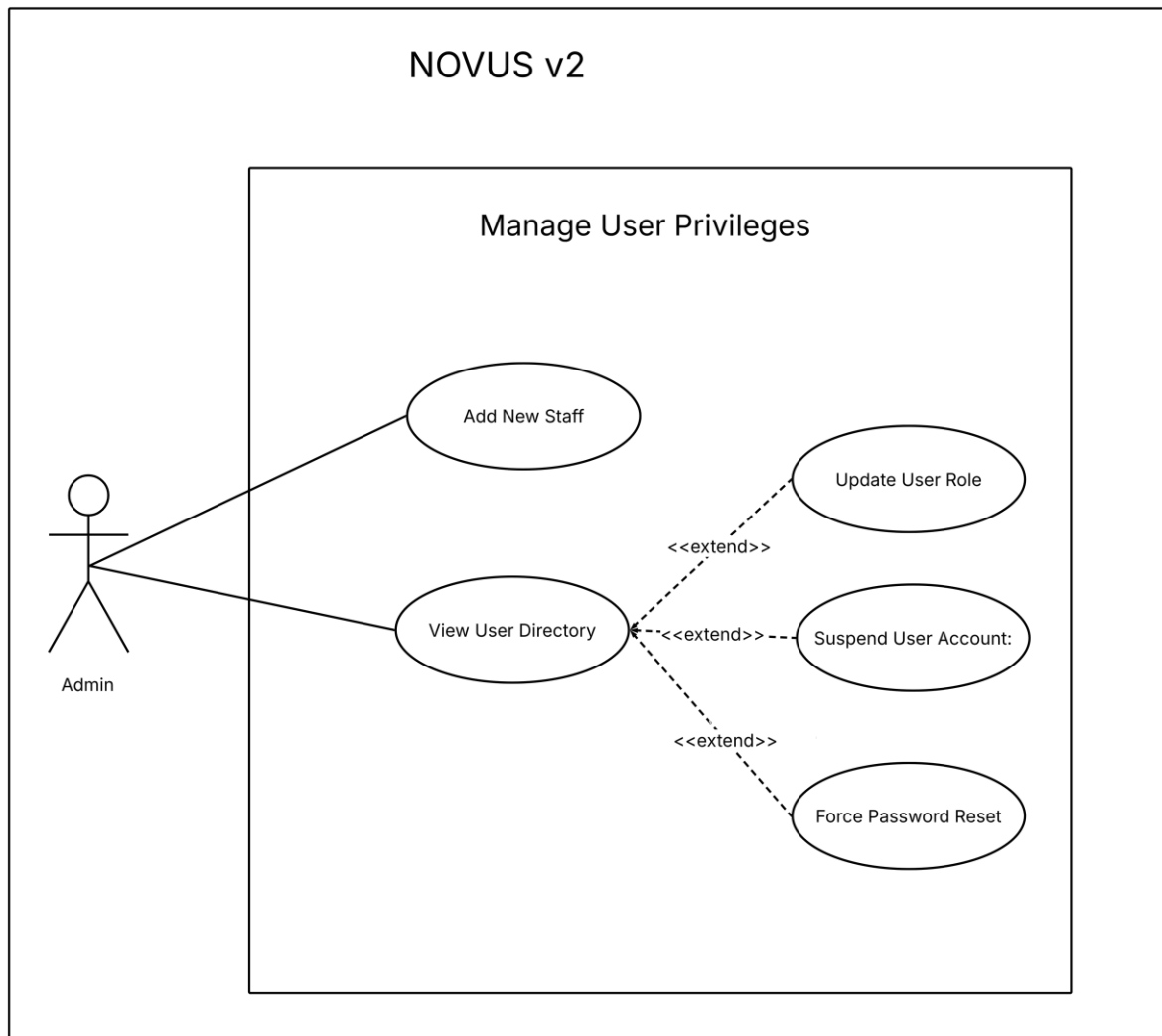
### UC-4.3: Toggle Maintenance Mode

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-4.3                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Author</b>                   | Novus Development Team                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose</b>                  | Allow the Admin to place the platform into maintenance mode, restricting reader access during updates.                                                                                                                                                                                                                                                                        |
| <b>Related Use Case</b>         | None                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Requirement Traceability</b> | PR-SET-03                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Priority</b>                 | Medium                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Preconditions</b>            | System is online<br>Admin is authenticated and logged in                                                                                                                                                                                                                                                                                                                      |
| <b>Postconditions</b>           | Platform maintenance state is toggled and reflected to visitors                                                                                                                                                                                                                                                                                                               |
| <b>Actors</b>                   | Admin                                                                                                                                                                                                                                                                                                                                                                         |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>Admin selects "Toggle Maintenance Mode."</li> <li>System prompts for confirmation and an optional custom maintenance message.</li> <li>Admin confirms.</li> <li>System enables maintenance mode, displaying the message to all non-Admin visitors.</li> <li>Admin later disables maintenance mode to restore normal access.</li> </ol> |

| USE CASE ELEMENT                                                        | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALTERNATE FLOW<br/>A1 (Scheduled<br/>Maintenance)</b>                | <ol style="list-style-type: none"> <li>1. The user enters a date/time.</li> <li>2. The system validates format and scheduling rules.</li> <li>3. The system detects that the value violates a constraint.</li> <li>4. The system rejects the value and explains the rule.</li> <li>5. The user enters a compliant future value.</li> <li>6. The system revalidates and continues.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>EXCEPTION FLOW<br/>E1 (Publishing Conflict)</b>                      | <ol style="list-style-type: none"> <li>1. The user confirms publication.</li> <li>2. The system validates publish readiness.</li> <li>3. The system starts the publication transaction and records the pre-publication state.</li> <li>4. The publishing service returns an error, timeout, or incomplete result.</li> <li>5. The system verifies what, if anything, reached the live environment.</li> <li>6. The system does not report full publication when verification is incomplete.</li> <li>7. The system records the failure and exposes controlled recovery/retry.</li> </ol>                                                                                                                                                                                                      |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization<br/>or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>MODULE 5: Manage User Privileges</p> <p>Actors: Admin</p> <p>UC-5.1: Add New User</p> <p>Use Case Name: Manage User Privileges_Add New User</p> |

### 3.5 Module 5: Manage User Privileges

**Figure 6. Manage User Privileges Use Case Diagram**



### UC-5.1: Add New User

| USE CASE ELEMENT         | SPECIFICATION                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-5.1                                                                                                                                                                                                                                                                      |
| Author                   | Novus Development Team                                                                                                                                                                                                                                                      |
| Purpose                  | Allow the Admin to manually create a new staff or reader account with an assigned role.                                                                                                                                                                                     |
| Related Use Case         | • None                                                                                                                                                                                                                                                                      |
| Requirement Traceability | PR-USR-01                                                                                                                                                                                                                                                                   |
| Priority                 | High                                                                                                                                                                                                                                                                        |
| Preconditions            | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Admin is authenticated and logged in</li> </ul>                                                                                                                                                        |
| Postconditions           | <ul style="list-style-type: none"> <li>• New user account is created with the assigned role</li> </ul>                                                                                                                                                                      |
| Actors                   | <ul style="list-style-type: none"> <li>• Admin</li> </ul>                                                                                                                                                                                                                   |
| BASIC FLOW               | <ol style="list-style-type: none"> <li>1. Admin selects “Add New Staff.”</li> <li>2. System displays a new-user form.</li> <li>3. Admin enters the name, email, and assigns a role (Editor, Writer, Publisher, Admin, etc.).</li> <li>4. Admin submits the form.</li> </ol> |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | 5. System creates the account and sends setup credentials to the user's email.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>ALTERNATE FLOW<br/>A1 (Bulk Import)</b>                          | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                |
| <b>EXCEPTION FLOW<br/>E1 (Duplicate Email)</b>                      | <ol style="list-style-type: none"> <li>1. The user submits the operation.</li> <li>2. The system checks unique identifiers against existing records.</li> <li>3. The system finds an existing matching record.</li> <li>4. The system prevents duplicate creation.</li> <li>5. The system displays a controlled message without exposing unrelated private information.</li> <li>6. The user chooses the permitted recovery action.</li> <li>7. The system continues only with a valid unique target.</li> </ol>                                                                                                                                                                                                                            |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-5.2: View User Directory<br/>Use Case Name: Manage User Privileges_View User Directory</p> |

### UC-5.2: View User Directory

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-5.2                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Author</b>                   | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose</b>                  | Allow the Admin to view a searchable list of all system users along with their roles and status.                                                                                                                                                                                                                                                                                                  |
| <b>Related Use Case</b>         | <ul style="list-style-type: none"> <li>• Extended by: Update User Role (UC-5.2.1), Suspend User Account (UC-5.2.2), Force Password Reset (UC-5.2.3)</li> </ul>                                                                                                                                                                                                                                    |
| <b>Requirement Traceability</b> | PR-USR-02                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Priority</b>                 | Medium                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Preconditions</b>            | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Admin is authenticated and logged in</li> </ul>                                                                                                                                                                                                                                                                              |
| <b>Postconditions</b>           | <ul style="list-style-type: none"> <li>• User directory is displayed</li> </ul>                                                                                                                                                                                                                                                                                                                   |
| <b>Actors</b>                   | <ul style="list-style-type: none"> <li>• Admin</li> </ul>                                                                                                                                                                                                                                                                                                                                         |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. Admin selects "View User Directory."</li> <li>2. System retrieves and displays all user accounts.</li> <li>3. Admin searches or filters by name, role, or status. From this listing, Admin may select a user record and optionally trigger Update User Role (UC-5.2.1), Suspend User Account (UC-5.2.2), or Force Password Reset (UC-5.2.3).</li> </ol> |

| USE CASE ELEMENT                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALTERNATE FLOW<br/>A1 (Export Directory)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                 |
| <b>EXCEPTION FLOW<br/>E1 (Load Failure)</b>     | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>UC-5.2.1: Update User Role</p> <p>Use Case Name: Manage User Privileges_Update User Role</p> |

### UC-5.2.1: Update User Role

| USE CASE ELEMENT                              | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                            | UC-5.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Author</b>                                 | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose</b>                                | Allow the Admin to change a user's assigned role and permission level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Related Use Case</b>                       | <ul style="list-style-type: none"> <li>• Extends: View User Directory (UC-5.2)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Requirement Traceability</b>               | PR-USR-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Priority</b>                               | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Preconditions</b>                          | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Admin is authenticated and logged in, and is currently viewing the User Directory (UC-5.2)</li> <li>• Target user account exists in the directory listing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Postconditions</b>                         | <ul style="list-style-type: none"> <li>• User's role and permissions are updated</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Actors</b>                                 | <ul style="list-style-type: none"> <li>• Admin</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>BASIC FLOW</b>                             | <ol style="list-style-type: none"> <li>1. While viewing the User Directory (UC-5.2), Admin selects a target user record and triggers "Update User Role."</li> <li>2. System displays available roles.</li> <li>3. Admin selects the new role.</li> <li>4. Admin confirms.</li> <li>5. System updates the user's role, applies the permission changes immediately, and returns control to the User Directory (UC-5.2).</li> </ol>                                                                                                                                                                                                             |
| <b>ALTERNATE FLOW<br/>A1 (Multiple Roles)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |
| <b>EXCEPTION FLOW</b>                         | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| USE CASE ELEMENT                  | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E1 (Last Admin Protection)</b> | <ol style="list-style-type: none"> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>UC-5.2.2: Suspend User Account<br/>Use Case Name: Manage User Privileges_Suspend User Account</p> |

### UC-5.2.2: Suspend User Account

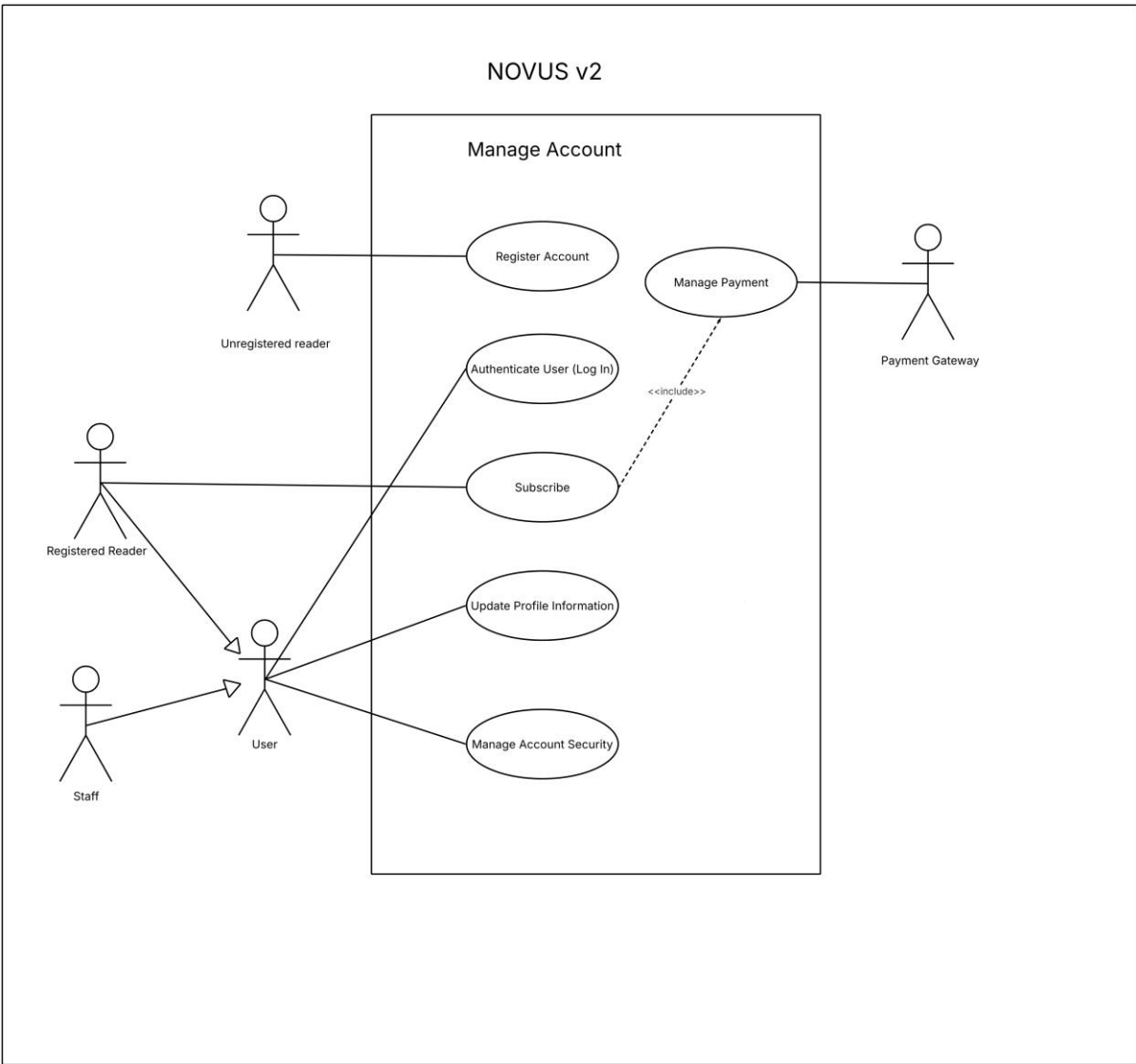
| USE CASE ELEMENT                                          | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                                        | UC-5.2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Author</b>                                             | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose</b>                                            | Allow the Admin to temporarily disable a user's access to the platform.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Related Use Case</b>                                   | <ul style="list-style-type: none"> <li>Extends: View User Directory (UC-5.2)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Requirement Traceability</b>                           | PR-USR-04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Priority</b>                                           | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Preconditions</b>                                      | <ul style="list-style-type: none"> <li>System is online</li> <li>Admin is authenticated and logged in, and is currently viewing the User Directory (UC-5.2)</li> <li>Target user account exists in the directory listing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Postconditions</b>                                     | <ul style="list-style-type: none"> <li>User account is suspended and active sessions are revoked</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Actors</b>                                             | <ul style="list-style-type: none"> <li>Admin</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>BASIC FLOW</b>                                         | <ol style="list-style-type: none"> <li>While viewing the User Directory (UC-5.2), Admin selects a target user record and triggers "Suspend User Account."</li> <li>System prompts for a reason and, optionally, a duration.</li> <li>Admin confirms.</li> <li>System suspends the account, revokes active sessions, and returns control to the User Directory (UC-5.2).</li> </ol>                                                                                                                                                                                                                                                                                                                                           |
| <b>ALTERNATE FLOW<br/>A1 (Reinstate Account)</b>          | <ol style="list-style-type: none"> <li>The user requests an operation requiring active entitlement/account status.</li> <li>The system retrieves the authoritative status.</li> <li>The system determines the account or entitlement is inactive, expired, or suspended.</li> <li>The system blocks the protected operation.</li> <li>The system displays the permitted recovery action.</li> <li>The system records the denied attempt when required.</li> </ol>                                                                                                                                                                                                                                                            |
| <b>EXCEPTION FLOW<br/>E1 (Self-Suspension Protection)</b> | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>UC-5.2.3: Force Password Reset<br/>Use Case Name: Manage User Privileges_Force Password Reset</p> |

### UC-5.2.3: Force Password Reset

| USE CASE ELEMENT                           | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                                | UC-5.2.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Author                                     | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Purpose                                    | Allow the Admin to require a user to reset their password on next login, for example after a security concern.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Related Use Case                           | <ul style="list-style-type: none"> <li>Extends: View User Directory (UC-5.2)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Requirement Traceability                   | PR-USR-05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Priority                                   | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Preconditions                              | <ul style="list-style-type: none"> <li>System is online</li> <li>Admin is authenticated and logged in, and is currently viewing the User Directory (UC-5.2)</li> <li>Target user account exists in the directory listing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Postconditions                             | <ul style="list-style-type: none"> <li>Account is flagged for mandatory password reset</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Actors                                     | <ul style="list-style-type: none"> <li>Admin</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| BASIC FLOW                                 | <ol style="list-style-type: none"> <li>While viewing the User Directory (UC-5.2), Admin selects a target user record and triggers “Force Password Reset.”</li> <li>System asks the Admin to confirm the action.</li> <li>System invalidates the current password and flags the account for mandatory reset.</li> <li>System emails the user a reset link and returns control to the User Directory (UC-5.2).</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ALTERNATE FLOW<br>A1 (Bulk Reset)          | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                                                                                                                                                                          |
| EXCEPTION FLOW<br>E1 (Undeliverable Email) | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>MODULE 6: Manage Account</p> <p>Actors: Unregistered Reader; Registered Reader; Staff; User (generalized parent of Registered Reader and Staff); Payment Gateway (external)</p> <p>UC-6.1: Register Account</p> <p>Use Case Name: Manage Account_Register Account</p> |

## 3.6 Module 6: Manage Account

Figure 7. Manage Account Use Case Diagram



### UC-6.1: Register Account

| USE CASE ELEMENT         | SPECIFICATION                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-6.1                                                                                          |
| Author                   | Novus Development Team                                                                          |
| Purpose                  | Allow an Unregistered Reader to create a new reader account.                                    |
| Related Use Case         | None                                                                                            |
| Requirement Traceability | PR-ACC-01                                                                                       |
| Priority                 | High                                                                                            |
| Preconditions            | System is online<br>Unregistered Reader has navigated to the application's URL                  |
| Postconditions           | A new Registered Reader account is created and able to log in                                   |
| Actors                   | Unregistered Reader                                                                             |
| BASIC FLOW               | 1. Unregistered Reader clicks "Log In or Sign Up."<br>2. System displays the registration form. |

| USE CASE ELEMENT                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <ol style="list-style-type: none"> <li>3. Reader enters their name, email, and password.</li> <li>4. Reader submits the form.</li> <li>5. System validates uniqueness of the email and creates the account.</li> <li>6. System sends a verification email.</li> <li>7. Reader verifies the email to activate the account.</li> </ol>                                                                                                                                                                                                                                                                                                         |
| <b>ALTERNATE FLOW<br/>A1 (SSO Registration)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |
| <b>EXCEPTION FLOW<br/>E1 (Duplicate Email)</b>  | <ol style="list-style-type: none"> <li>1. The user submits the operation.</li> <li>2. The system checks unique identifiers against existing records.</li> <li>3. The system finds an existing matching record.</li> <li>4. The system prevents duplicate creation.</li> <li>5. The system displays a controlled message without exposing unrelated private information.</li> <li>6. The user chooses the permitted recovery action.</li> <li>7. The system continues only with a valid unique target.</li> </ol> <p>UC-6.2: Authenticate User (Log In)</p> <p>Use Case Name: Manage Account_Authenticate User (Log In)</p>                   |

### UC-6.2: Authenticate User (Log In)

| USE CASE ELEMENT                         | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                       | UC-6.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Author</b>                            | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose</b>                           | Allow a registered User, whether a Reader or Staff member, to log in to their account.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Related Use Case</b>                  | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Requirement Traceability</b>          | PR-ACC-02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Priority</b>                          | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Preconditions</b>                     | System is online<br>User has a registered, active account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Postconditions</b>                    | An authenticated session is created for the User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Actors</b>                            | User (Registered Reader, Staff)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>BASIC FLOW</b>                        | <ol style="list-style-type: none"> <li>1. User navigates to the login page.</li> <li>2. User enters their email/username and password.</li> <li>3. User submits the form.</li> <li>4. System validates the credentials.</li> <li>5. System creates an authenticated session and redirects the user to their dashboard.</li> </ol>                                                                                                                                                                                                                 |
| <b>ALTERNATE FLOW<br/>A1 (SSO Login)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> </ol> |

| USE CASE ELEMENT                                   | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | 7. The user continues from the appropriate point after the condition is resolved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>EXCEPTION FLOW<br/>E1 (Invalid Credentials)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                         |
| <b>EXCEPTION FLOW<br/>E2 (Suspended Account)</b>   | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>UC-6.3: Subscribe</p> <p>Use Case Name: Manage Account_Subscribe</p> |

### UC-6.3: Subscribe

| USE CASE ELEMENT                                     | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                                   | UC-6.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Author</b>                                        | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose</b>                                       | Allow a Registered Reader to purchase a premium subscription plan, upgrading their access to Subscriber status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Related Use Case</b>                              | Includes: Manage Payment module — Subscribe to Premium Plan (UC-9.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Requirement Traceability</b>                      | PR-ACC-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Priority</b>                                      | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Preconditions</b>                                 | System is online<br>Registered Reader is authenticated and logged in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Postconditions</b>                                | On successful payment, the Reader's account is upgraded to Subscriber status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Actors</b>                                        | Registered Reader<br>Payment Gateway (secondary, via include)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BASIC FLOW</b>                                    | <ol style="list-style-type: none"> <li>1. Registered Reader selects "Subscribe."</li> <li>2. System displays the available subscription plans.</li> <li>3. Reader selects a plan.</li> <li>4. System includes the Manage Payment module to process the transaction (UC-9.1).</li> <li>5. Upon successful payment, system upgrades the Reader's account to Subscriber status.</li> <li>6. System confirms the upgrade to the user.</li> </ol>                                                                                                                                                                                                 |
| <b>ALTERNATE FLOW<br/>A1 (Switch Plan Selection)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALTERNATE FLOW<br/>A2 (Existing Active Subscription)</b>         | <ol style="list-style-type: none"> <li>1. The user opens the dashboard.</li> <li>2. The system verifies session and permissions.</li> <li>3. The system queries current records.</li> <li>4. If no records match, the system displays the configured empty state.</li> <li>5. If retrieval fails, the system displays a controlled error and retry option.</li> <li>6. The system logs the failure when appropriate.</li> <li>7. The user retries or changes the filter.</li> </ol>                                                                                                                                                                                                                                                                  |
| <b>ALTERNATE FLOW<br/>A3 (Checkout Cancellation)</b>                | <ol style="list-style-type: none"> <li>1. The user chooses to cancel, discard, or leave the operation.</li> <li>2. The system checks for unsaved changes or a pending transaction.</li> <li>3. The system requests confirmation when the action could lose work or affect billing.</li> <li>4. The user confirms cancellation.</li> <li>5. The system discards only uncommitted changes or safely terminates the pending operation.</li> <li>6. The system restores the last confirmed state and returns the user to the appropriate screen.</li> <li>7. The system does not create an unintended version, state change, or entitlement.</li> </ol>                                                                                                  |
| <b>EXCEPTION FLOW<br/>E1 (Payment Failure)</b>                      | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                         |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-6.4: Update Profile Information</p> <p>Use Case Name: Manage Account_Update Profile Information</p> |

### UC-6.4: Update Profile Information

| USE CASE ELEMENT                | SPECIFICATION                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-6.4                                                                                             |
| <b>Author</b>                   | Novus Development Team                                                                             |
| <b>Purpose</b>                  | Allow a User to update their personal profile details, such as name, avatar, bio, and preferences. |
| <b>Related Use Case</b>         | None                                                                                               |
| <b>Requirement Traceability</b> | PR-ACC-04                                                                                          |
| <b>Priority</b>                 | Low                                                                                                |
| <b>Preconditions</b>            | System is online<br>User is authenticated and logged in                                            |
| <b>Postconditions</b>           | Profile information is updated                                                                     |
| <b>Actors</b>                   | User (Registered Reader, Staff)                                                                    |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. User navigates to Account Settings.</li> </ol>           |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <ol style="list-style-type: none"> <li>2. User selects "Update Profile Information."</li> <li>3. System displays the current profile form.</li> <li>4. User edits the desired fields.</li> <li>5. User saves.</li> <li>6. System validates and stores the updated profile.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>ALTERNATE FLOW<br/>A1 (Photo Upload)</b>                         | <ol style="list-style-type: none"> <li>1. The user selects a file and begins upload.</li> <li>2. The system validates role, target assignment, file type, file size, and upload rules.</li> <li>3. The system rejects invalid files before finalization.</li> <li>4. For a valid file, the system transfers it to controlled storage.</li> <li>5. The system verifies transfer completion and file integrity.</li> <li>6. The system creates a new version and preserves prior versions.</li> <li>7. If interrupted, the system prevents the incomplete file from becoming active and marks it failed or resumable.</li> <li>8. The user retries or resumes after correcting the problem.</li> </ol>                                           |
| <b>EXCEPTION FLOW<br/>E1 (Invalid Input)</b>                        | <ol style="list-style-type: none"> <li>1. The user enters or confirms information.</li> <li>2. The system validates format, range, required fields, and business rules.</li> <li>3. The system identifies invalid values.</li> <li>4. The system highlights the affected fields and required corrections.</li> <li>5. The system prevents persistence of invalid data.</li> <li>6. The user corrects the values.</li> <li>7. The system validates again and resumes the flow.</li> </ol>                                                                                                                                                                                                                                                       |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-6.5: Manage Account Security</p> <p>Use Case Name: Manage Account_Manage Account Security</p> |

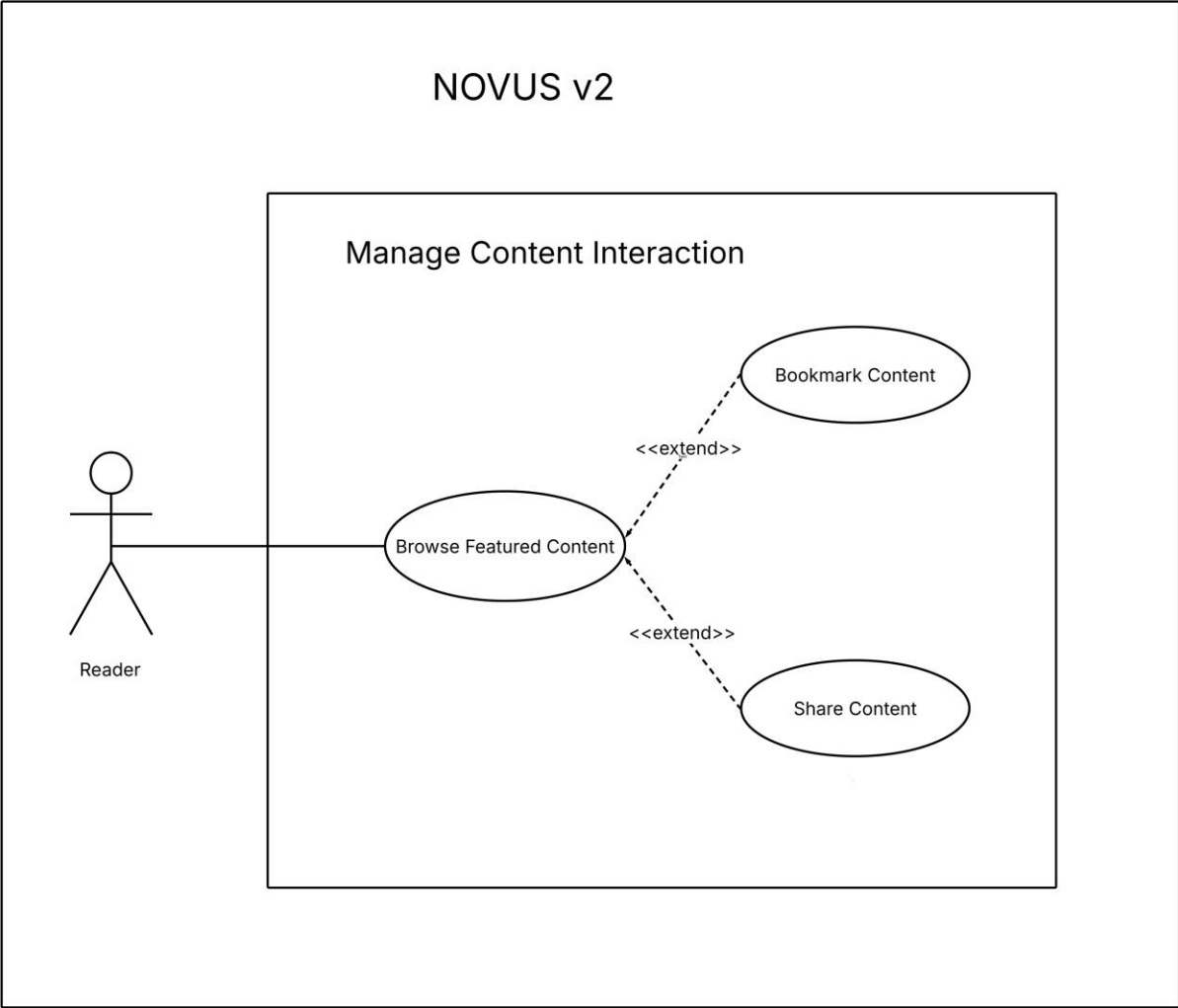
### UC-6.5: Manage Account Security

| USE CASE ELEMENT                | SPECIFICATION                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-6.5                                                                                                |
| <b>Author</b>                   | Novus Development Team                                                                                |
| <b>Purpose</b>                  | Allow a User to change their password and manage security settings such as two-factor authentication. |
| <b>Related Use Case</b>         | None                                                                                                  |
| <b>Requirement Traceability</b> | PR-ACC-05                                                                                             |
| <b>Priority</b>                 | High                                                                                                  |
| <b>Preconditions</b>            | System is online<br>User is authenticated and logged in                                               |
| <b>Postconditions</b>           | Security settings are updated; other active sessions are invalidated if the password changed          |
| <b>Actors</b>                   | User (Registered Reader, Staff)                                                                       |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. User navigates to Account Settings.</li> </ol>              |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <ol style="list-style-type: none"><li>2. User selects "Manage Account Security."</li><li>3. System displays security options: change password, enable two-factor authentication, view active sessions.</li><li>4. User updates the desired settings.</li><li>5. User confirms the changes.</li><li>6. System applies the changes and, if the password changed, invalidates other active sessions.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>ALTERNATE FLOW<br/>A1 (Enable 2FA)</b>                           | <ol style="list-style-type: none"><li>1. The user reaches the condition described by this flow.</li><li>2. The system detects the condition during the relevant basic-flow step.</li><li>3. The system validates the current record, permission, and business rules.</li><li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li><li>5. The system explains the resulting state and required next action.</li><li>6. The system records the event when audit/security rules require it.</li><li>7. The user continues from the appropriate point after the condition is resolved.</li></ol>                                                                                                                                                                                                     |
| <b>EXCEPTION FLOW<br/>E1 (Incorrect Current Password)</b>           | <ol style="list-style-type: none"><li>1. The user reaches the condition described by this flow.</li><li>2. The system detects the condition during the relevant basic-flow step.</li><li>3. The system validates the current record, permission, and business rules.</li><li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li><li>5. The system explains the resulting state and required next action.</li><li>6. The system records the event when audit/security rules require it.</li><li>7. The user continues from the appropriate point after the condition is resolved.</li></ol>                                                                                                                                                                                                     |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"><li>1. The user initiates the protected operation.</li><li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li><li>3. The system detects an expired session or missing permission.</li><li>4. The system blocks the state change and does not partially commit it.</li><li>5. The system records the denied attempt when required by security policy.</li><li>6. The system redirects to authentication or displays access denied.</li><li>7. After authorization is restored, the user may retry.</li></ol> <p>MODULE 7: Manage Content Interaction</p> <p>Actors: Reader (Registered Reader)</p> <p>UC-7.1: Browse Featured Content</p> <p>Use Case Name: Manage Content Interaction_Browse Featured Content</p> |

3.7 Module 7: Manage Content Interaction

Figure 8. Manage Content Interaction Use Case Diagram



**UC-7.1: Browse Featured Content**

| USE CASE ELEMENT         | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-7.1                                                                                                                                                                                                                                                                                                                                                                              |
| Author                   | Novus Development Team                                                                                                                                                                                                                                                                                                                                                              |
| Purpose                  | Allow a Reader to browse curated and featured articles and issues on the platform.                                                                                                                                                                                                                                                                                                  |
| Related Use Case         | <ul style="list-style-type: none"><li>Extended by: Bookmark Content (UC-7.1.1), Share Content (UC-7.1.2)</li></ul>                                                                                                                                                                                                                                                                  |
| Requirement Traceability | PR-INT-01                                                                                                                                                                                                                                                                                                                                                                           |
| Priority                 | Medium                                                                                                                                                                                                                                                                                                                                                                              |
| Preconditions            | <ul style="list-style-type: none"><li>System is online</li><li>Reader has access to the homepage or featured section</li></ul>                                                                                                                                                                                                                                                      |
| Postconditions           | <ul style="list-style-type: none"><li>Featured content is displayed to the Reader</li></ul>                                                                                                                                                                                                                                                                                         |
| Actors                   | <ul style="list-style-type: none"><li>Reader</li></ul>                                                                                                                                                                                                                                                                                                                              |
| BASIC FLOW               | <ol style="list-style-type: none"><li>Reader navigates to the homepage or featured section.</li><li>System retrieves and displays featured content.</li><li>Reader scrolls or browses through the featured items.</li><li>Reader selects an article to read; while viewing it, the Reader may optionally trigger Bookmark Content (UC-7.1.1) or Share Content (UC-7.1.2).</li></ol> |

| USE CASE ELEMENT                                   | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ALTERNATE FLOW<br/>A1 (Category Filter)</b>     | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |
| <b>EXCEPTION FLOW<br/>E1 (No Featured Content)</b> | <ol style="list-style-type: none"> <li>1. The user requests the selected list/report.</li> <li>2. The system queries the current data set.</li> <li>3. The system finds no matching records.</li> <li>4. The system displays the empty state or configured fallback.</li> <li>5. The system does not present fabricated or stale data.</li> <li>6. The user adjusts filters or returns to another section.</li> </ol> <p>UC-7.1.1: Bookmark Content</p> <p>Use Case Name: Manage Content Interaction_Bookmark Content</p>                                                                                                                    |

### UC-7.1.1: Bookmark Content

| USE CASE ELEMENT                               | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                             | UC-7.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Author</b>                                  | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose</b>                                 | Allow a Reader to save an article for later reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Related Use Case</b>                        | <ul style="list-style-type: none"> <li>• Extends: Browse Featured Content (UC-7.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Requirement Traceability</b>                | PR-INT-02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Priority</b>                                | Low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Preconditions</b>                           | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Reader is authenticated and logged in</li> <li>• Reader is viewing an article reached through Browse Featured Content (UC-7.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Postconditions</b>                          | <ul style="list-style-type: none"> <li>• Article is added to the Reader's bookmark list</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Actors</b>                                  | <ul style="list-style-type: none"> <li>• Reader</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>BASIC FLOW</b>                              | <ol style="list-style-type: none"> <li>1. While viewing an article reached through Browse Featured Content (UC-7.1), Reader selects "Bookmark."</li> <li>2. System adds the article to the Reader's bookmark list.</li> <li>3. System confirms with a visual indicator and returns control to Browse Featured Content (UC-7.1).</li> </ol>                                                                                                                                                                                                                                                                                                   |
| <b>ALTERNATE FLOW<br/>A1 (Remove Bookmark)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> |
| <b>EXCEPTION FLOW<br/>E1 (Not Logged In)</b>   | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> </ol>                                                                                                                                                                           |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | 6. The system records the event when audit/security rules require it.<br>7. The user continues from the appropriate point after the condition is resolved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | 1. The user initiates the protected operation.<br>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.<br>3. The system detects an expired session or missing permission.<br>4. The system blocks the state change and does not partially commit it.<br>5. The system records the denied attempt when required by security policy.<br>6. The system redirects to authentication or displays access denied.<br>7. After authorization is restored, the user may retry.<br>UC-7.1.2: Share Content<br>Use Case Name: Manage Content Interaction_Share Content |

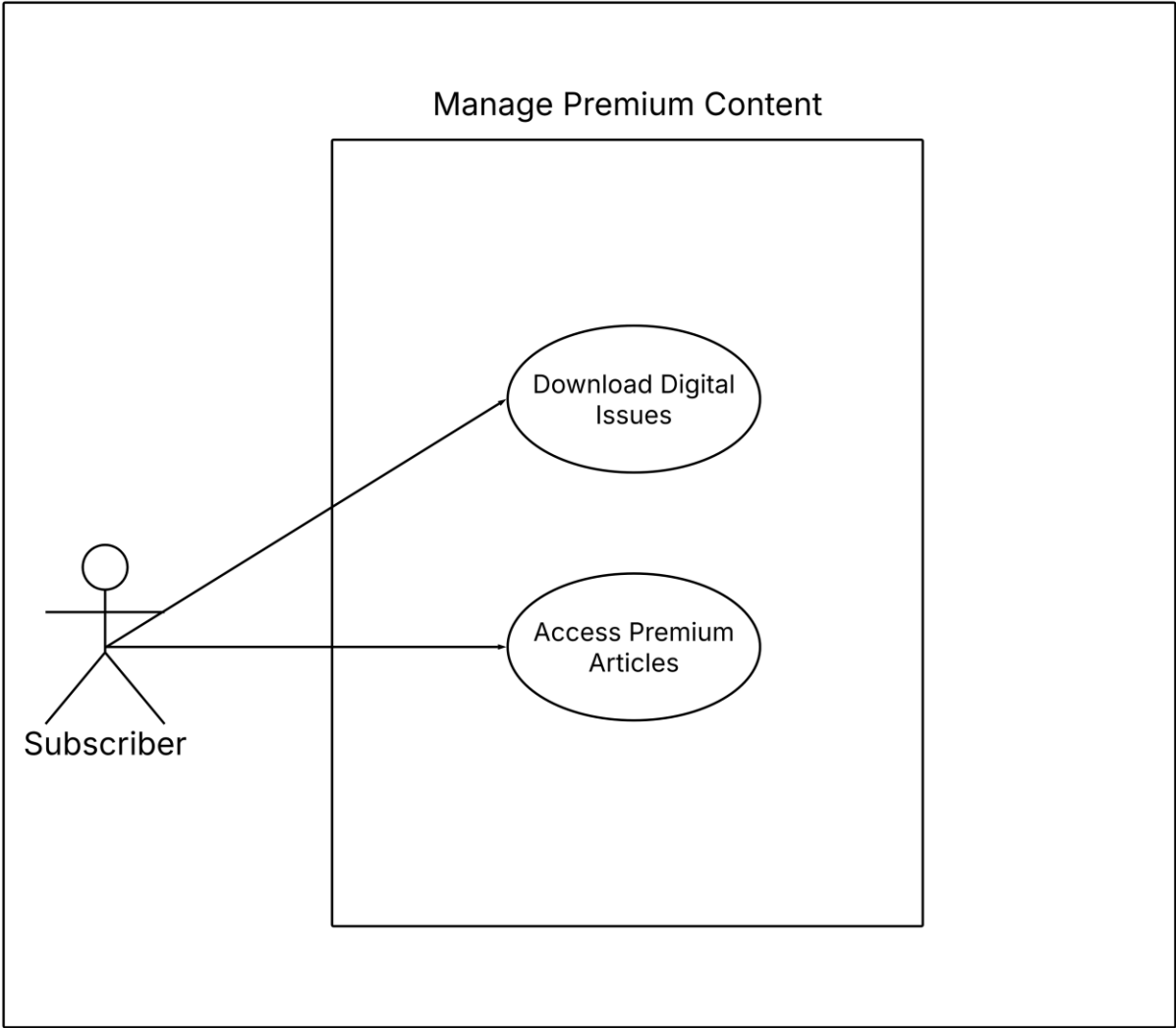
### UC-7.1.2: Share Content

| USE CASE ELEMENT                                   | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>                                 | UC-7.1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Author</b>                                      | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose</b>                                     | Allow a Reader to share an article link via social media or a direct link copy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Related Use Case</b>                            | <ul style="list-style-type: none"> <li>Extends: Browse Featured Content (UC-7.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Requirement Traceability</b>                    | PR-INT-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Priority</b>                                    | Low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>                               | <ul style="list-style-type: none"> <li>System is online</li> <li>Reader is viewing an article reached through Browse Featured Content (UC-7.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Postconditions</b>                              | <ul style="list-style-type: none"> <li>A shareable link is generated and the share action is completed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Actors</b>                                      | <ul style="list-style-type: none"> <li>Reader</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>BASIC FLOW</b>                                  | 1. While viewing an article reached through Browse Featured Content (UC-7.1), Reader selects “Share.”<br>2. System displays sharing options (social platforms, copy link).<br>3. Reader selects a channel.<br>4. System generates the shareable link, completes the share action, and returns control to Browse Featured Content (UC-7.1).                                                                                                                                                                                                             |
| <b>ALTERNATE FLOW<br/>A1 (Copy Link)</b>           | 1. The user reaches the condition described by this flow.<br>2. The system detects the condition during the relevant basic-flow step.<br>3. The system validates the current record, permission, and business rules.<br>4. The system performs the permitted alternate handling or blocks the invalid operation.<br>5. The system explains the resulting state and required next action.<br>6. The system records the event when audit/security rules require it.<br>7. The user continues from the appropriate point after the condition is resolved. |
| <b>EXCEPTION FLOW<br/>E1 (Service Unreachable)</b> | 1. The user requests the external-service operation.<br>2. The system validates the endpoint and configured credentials.<br>3. The system attempts the connection or transaction.<br>4. The external service cannot be reached.<br>5. The system does not save unverified settings or treat the transaction as successful.<br>6. The system displays a controlled error and records diagnostics without exposing secrets.<br>7. The user retries after correction or service recovery.                                                                 |

| USE CASE ELEMENT | SPECIFICATION                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | MODULE 8: Manage Premium Content<br>Actors: Subscriber<br>UC-8.1: Download Digital Issues<br>Use Case Name: Manage Premium Content_Download Digital Issues |

### 3.8 Module 8: Manage Premium Content

Figure 9. Manage Premium Content Use Case Diagram



#### UC-8.1: Download Digital Issues

| USE CASE ELEMENT         | SPECIFICATION                                                            |
|--------------------------|--------------------------------------------------------------------------|
| Use Case ID              | UC-8.1                                                                   |
| Author                   | Novus Development Team                                                   |
| Purpose                  | Allow a Subscriber to download a full digital issue for offline reading. |
| Related Use Case         | None                                                                     |
| Requirement Traceability | PR-PRM-01                                                                |

| USE CASE ELEMENT                            | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority                                    | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Preconditions                               | System is online<br>Subscriber is authenticated and logged in<br>Subscriber has an active premium subscription                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Postconditions                              | The digital issue file is downloaded to the Subscriber's device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Actors                                      | Subscriber                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| BASIC FLOW                                  | 1. Subscriber navigates to an available issue.<br>2. Subscriber selects "Download Digital Issue."<br>3. System verifies the Subscriber's active premium status.<br>4. System generates or serves the downloadable file.<br>5. Subscriber's device saves the file.                                                                                                                                                                                                                                                                                                                                                                                           |
| ALTERNATE FLOW<br>A1 (Format Choice)        | 1. The user selects a file and begins upload.<br>2. The system validates role, target assignment, file type, file size, and upload rules.<br>3. The system rejects invalid files before finalization.<br>4. For a valid file, the system transfers it to controlled storage.<br>5. The system verifies transfer completion and file integrity.<br>6. The system creates a new version and preserves prior versions.<br>7. If interrupted, the system prevents the incomplete file from becoming active and marks it failed or resumable.<br>8. The user retries or resumes after correcting the problem.                                                    |
| EXCEPTION FLOW<br>E1 (Expired Subscription) | 1. The user reaches the condition described by this flow.<br>2. The system detects the condition during the relevant basic-flow step.<br>3. The system validates the current record, permission, and business rules.<br>4. The system performs the permitted alternate handling or blocks the invalid operation.<br>5. The system explains the resulting state and required next action.<br>6. The system records the event when audit/security rules require it.<br>7. The user continues from the appropriate point after the condition is resolved.<br>UC-8.2: Access Premium Articles<br>Use Case Name: Manage Premium Content _Access Premium Articles |

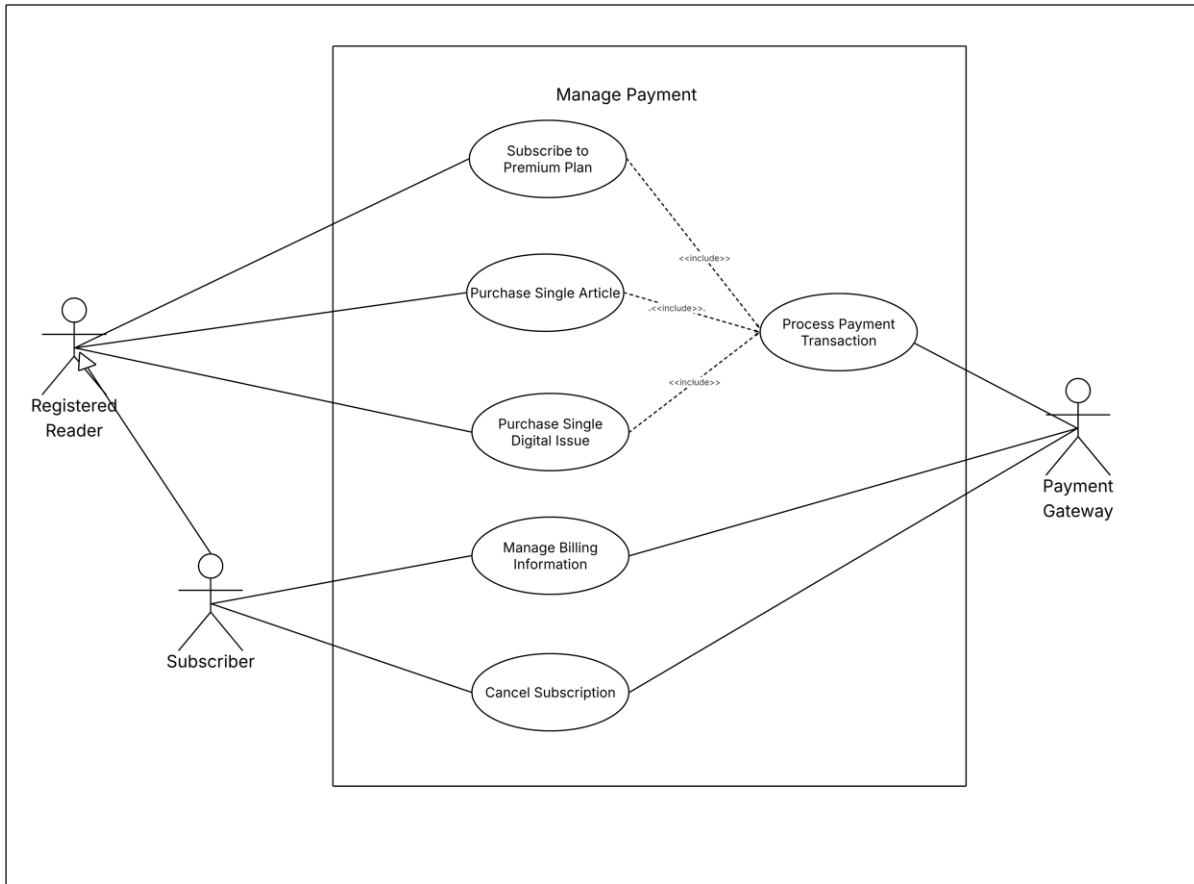
UC-8.2: Access Premium Articles

| USE CASE ELEMENT         | SPECIFICATION                                                              |
|--------------------------|----------------------------------------------------------------------------|
| Use Case ID              | UC-8.2                                                                     |
| Author                   | Novus Development Team                                                     |
| Purpose                  | Allow a Subscriber to view the full content of paywalled premium articles. |
| Related Use Case         | None                                                                       |
| Requirement Traceability | PR-PRM-02                                                                  |
| Priority                 | High                                                                       |
| Preconditions            | System is online<br>Subscriber is authenticated and logged in              |
| Postconditions           | Full premium article content is rendered for the Subscriber                |
| Actors                   | Subscriber                                                                 |
| BASIC FLOW               | 1. Subscriber selects a premium-marked article.                            |

| USE CASE ELEMENT                                           | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | <ol style="list-style-type: none"> <li>2. System verifies the Subscriber's active subscription status.</li> <li>3. System renders the full article content, bypassing the paywall preview.</li> <li>4. Subscriber reads the article.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>ALTERNATE FLOW</b><br><b>A1 (Single Article Access)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                                                                                                                                                       |
| <b>EXCEPTION FLOW</b><br><b>E1 (Inactive Subscription)</b> | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol> <p>MODULE 9: Manage Payment</p> <p>Actors: Registered Reader; Subscriber (generalized from Registered Reader); Payment Gateway (external)</p> <p>UC-9.1: Subscribe to Premium Plan</p> <p>Use Case Name: Manage Payment_Subscribe to Premium Plan</p> |

### 3.9 Module 9: Manage Payment

**Figure 10. Manage Payment Use Case Diagram**



### UC-9.1: Subscribe to Premium Plan

| USE CASE ELEMENT                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                         | UC-9.1                                                                                                                                                                                                                                                                                                                                                                                 |
| Author                              | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                 |
| Purpose                             | Process a Registered Reader's purchase of a recurring premium subscription plan via the Payment Gateway.                                                                                                                                                                                                                                                                               |
| Related Use Case                    | <ul style="list-style-type: none"> <li>Included by: Subscribe (UC-6.3)</li> <li>Includes: Process Payment Transaction (UC-9.1.1)</li> </ul>                                                                                                                                                                                                                                            |
| Requirement Traceability            | PR-PAY-01                                                                                                                                                                                                                                                                                                                                                                              |
| Priority                            | High                                                                                                                                                                                                                                                                                                                                                                                   |
| Preconditions                       | <ul style="list-style-type: none"> <li>System is online</li> <li>Registered Reader is authenticated and logged in</li> <li>Payment Gateway service is reachable</li> </ul>                                                                                                                                                                                                             |
| Postconditions                      | <ul style="list-style-type: none"> <li>Subscriber tier is activated on the Reader's account and the billing cycle is recorded</li> </ul>                                                                                                                                                                                                                                               |
| Actors                              | <ul style="list-style-type: none"> <li>Registered Reader</li> <li>Payment Gateway (secondary, via include: UC-9.1.1)</li> </ul>                                                                                                                                                                                                                                                        |
| BASIC FLOW                          | <ol style="list-style-type: none"> <li>Registered Reader selects a subscription plan and confirms intent to subscribe.</li> <li>System includes Process Payment Transaction (UC-9.1.1) to capture and process the payment.</li> <li>On receiving a success token from UC-9.1.1, System activates the Subscriber tier on the Reader's account and records the billing cycle.</li> </ol> |
| ALTERNATE FLOW<br>A1 (Saved Payment | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> </ol>                                                                                                                                                                                                |

| USE CASE ELEMENT                                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Method)</b>                                                  | <ol style="list-style-type: none"> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                                                                                                                                                                  |
| <b>EXCEPTION FLOW E1 (Transaction Declined)</b>                 | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                             |
| <b>EXCEPTION FLOW E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-9.1.1: Process Payment Transaction</p> <p>Use Case Name: Manage Payment_Process Payment Transaction</p> |

### UC-9.1.1: Process Payment Transaction

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-9.1.1                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Author</b>                   | Novus Development Team                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Purpose</b>                  | Serve as the central payment-processing engine that connects to the Payment Gateway, transmits transaction details, and returns a success or fail token to the invoking use case.                                                                                                                                                                                                                                                   |
| <b>Related Use Case</b>         | <ul style="list-style-type: none"> <li>• Included by: Subscribe to Premium Plan (UC-9.1), Purchase Single Article (UC-9.2), Purchase Single Digital Issue (UC-9.3)</li> </ul>                                                                                                                                                                                                                                                       |
| <b>Requirement Traceability</b> | PR-PAY-06                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Priority</b>                 | High                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Preconditions</b>            | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Payment Gateway service is reachable</li> <li>• The invoking use case has supplied the transaction amount and context</li> </ul>                                                                                                                                                                                                                               |
| <b>Postconditions</b>           | <ul style="list-style-type: none"> <li>• Transaction is processed by the Payment Gateway and a success or fail token, along with a transaction reference, is returned to the invoking use case</li> </ul>                                                                                                                                                                                                                           |
| <b>Actors</b>                   | <ul style="list-style-type: none"> <li>• Registered Reader (indirect, via invoking use case)</li> <li>• Payment Gateway</li> </ul>                                                                                                                                                                                                                                                                                                  |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. Invoking use case (UC-9.1, UC-9.2, or UC-9.3) triggers Process Payment Transaction and passes the transaction amount and context.</li> <li>2. System connects to the Payment Gateway.</li> <li>3. Reader enters or confirms payment details.</li> <li>4. Payment Gateway processes the transaction.</li> <li>5. Payment Gateway returns a success or fail token to the System.</li> </ol> |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | 6. System returns the token and transaction reference to the invoking use case.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>ALTERNATE FLOW<br/>A1 (Saved Payment Method)</b>                 | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                   |
| <b>EXCEPTION FLOW<br/>E1 (Transaction Declined)</b>                 | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                   |
| <b>EXCEPTION FLOW<br/>E2 (Gateway Unreachable)</b>                  | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                   |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-9.2: Purchase Single Article</p> <p>Use Case Name: Manage Payment_Purchase Single Article</p> |

### UC-9.2: Purchase Single Article

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-9.2                                                                                                                           |
| <b>Author</b>                   | Novus Development Team                                                                                                           |
| <b>Purpose</b>                  | Process a Registered Reader's one-time purchase of a single premium article without a full subscription.                         |
| <b>Related Use Case</b>         | <ul style="list-style-type: none"> <li>• Includes: Process Payment Transaction (UC-9.1.1)</li> </ul>                             |
| <b>Requirement Traceability</b> | PR-PAY-02                                                                                                                        |
| <b>Priority</b>                 | Medium                                                                                                                           |
| <b>Preconditions</b>            | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Registered Reader is authenticated and logged in</li> </ul> |
| <b>Postconditions</b>           | <ul style="list-style-type: none"> <li>• Reader is granted permanent access to the purchased article</li> </ul>                  |
| <b>Actors</b>                   | <ul style="list-style-type: none"> <li>• Registered Reader</li> </ul>                                                            |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <ul style="list-style-type: none"> <li>Payment Gateway (secondary, via include: UC-9.1.1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>BASIC FLOW</b>                                                   | <ol style="list-style-type: none"> <li>Registered Reader selects a premium article and chooses “Purchase Single Article.”</li> <li>System displays the price and includes Process Payment Transaction (UC-9.1.1) to capture and process the payment.</li> <li>On receiving a success token from UC-9.1.1, System grants the Reader permanent access to that specific article.</li> </ol>                                                                                                                                                                                                                                                                                                                                              |
| <b>ALTERNATE FLOW<br/>A1 (Bundle Purchase)</b>                      | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                               |
| <b>EXCEPTION FLOW<br/>E1 (Transaction Declined)</b>                 | <ol style="list-style-type: none"> <li>The user reaches the condition described by this flow.</li> <li>The system detects the condition during the relevant basic-flow step.</li> <li>The system validates the current record, permission, and business rules.</li> <li>The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>The system explains the resulting state and required next action.</li> <li>The system records the event when audit/security rules require it.</li> <li>The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                               |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>The user initiates the protected operation.</li> <li>The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>The system detects an expired session or missing permission.</li> <li>The system blocks the state change and does not partially commit it.</li> <li>The system records the denied attempt when required by security policy.</li> <li>The system redirects to authentication or displays access denied.</li> <li>After authorization is restored, the user may retry.</li> </ol> <p>UC-9.3: Purchase Single Digital Issue</p> <p>Use Case Name: Manage Payment_Purchase Single Digital Issue</p> |

### UC-9.3: Purchase Single Digital Issue

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-9.3                                                                                                                          |
| <b>Author</b>                   | Novus Development Team                                                                                                          |
| <b>Purpose</b>                  | Process a Registered Reader’s one-time purchase of a single digital issue without a full subscription.                          |
| <b>Related Use Case</b>         | <ul style="list-style-type: none"> <li>Includes: Process Payment Transaction (UC-9.1.1)</li> </ul>                              |
| <b>Requirement Traceability</b> | PR-PAY-03                                                                                                                       |
| <b>Priority</b>                 | Medium                                                                                                                          |
| <b>Preconditions</b>            | <ul style="list-style-type: none"> <li>System is online</li> <li>Registered Reader is authenticated and logged in</li> </ul>    |
| <b>Postconditions</b>           | <ul style="list-style-type: none"> <li>Reader is granted access and download rights to the purchased issue</li> </ul>           |
| <b>Actors</b>                   | <ul style="list-style-type: none"> <li>Registered Reader</li> <li>Payment Gateway (secondary, via include: UC-9.1.1)</li> </ul> |

| USE CASE ELEMENT                                                | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BASIC FLOW</b>                                               | <ol style="list-style-type: none"> <li>1. Registered Reader selects an issue and chooses “Purchase Single Digital Issue.”</li> <li>2. System displays the price and includes Process Payment Transaction (UC-9.1.1) to capture and process the payment.</li> <li>3. On receiving a success token from UC-9.1.1, System grants the Reader access and download rights to that issue.</li> </ol>                                                                                                                                                                                                                                                                                                                                                        |
| <b>ALTERNATE FLOW A1 (Gift Issue)</b>                           | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                         |
| <b>EXCEPTION FLOW E1 (Transaction Declined)</b>                 | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                         |
| <b>EXCEPTION FLOW E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-9.4: Manage Billing Information</p> <p>Use Case Name: Manage Payment_Manage Billing Information</p> |

### UC-9.4: Manage Billing Information

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-9.4                                                                                                                                                      |
| <b>Author</b>                   | Novus Development Team                                                                                                                                      |
| <b>Purpose</b>                  | Allow a Subscriber to view and update their stored billing and payment details.                                                                             |
| <b>Related Use Case</b>         | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                                    |
| <b>Requirement Traceability</b> | PR-PAY-04                                                                                                                                                   |
| <b>Priority</b>                 | Medium                                                                                                                                                      |
| <b>Preconditions</b>            | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Subscriber is authenticated and logged in</li> </ul>                                   |
| <b>Postconditions</b>           | <ul style="list-style-type: none"> <li>• Billing information is updated and validated with the Payment Gateway</li> </ul>                                   |
| <b>Actors</b>                   | <ul style="list-style-type: none"> <li>• Subscriber</li> <li>• Payment Gateway</li> </ul>                                                                   |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. Subscriber navigates to Billing Settings.</li> <li>2. Subscriber selects “Manage Billing Information.”</li> </ol> |

| USE CASE ELEMENT                                                    | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                     | <ol style="list-style-type: none"> <li>3. System retrieves current billing details via the Payment Gateway (masked card info).</li> <li>4. Subscriber updates the payment method or billing address.</li> <li>5. System validates the update with the Payment Gateway.</li> <li>6. System confirms the update.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>ALTERNATE FLOW<br/>A1 (Backup Payment Method)</b>                | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                           |
| <b>EXCEPTION FLOW<br/>E1 (Method Rejected)</b>                      | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                           |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>UC-9.5: Cancel Subscription</p> <p>Use Case Name: Manage Payment_Cancel Subscription</p> |

### UC-9.5: Cancel Subscription

| USE CASE ELEMENT                | SPECIFICATION                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-9.5                                                                                                                                                                     |
| <b>Author</b>                   | Novus Development Team                                                                                                                                                     |
| <b>Purpose</b>                  | Allow a Subscriber to cancel their recurring premium subscription.                                                                                                         |
| <b>Related Use Case</b>         | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                                                   |
| <b>Requirement Traceability</b> | PR-PAY-05                                                                                                                                                                  |
| <b>Priority</b>                 | High                                                                                                                                                                       |
| <b>Preconditions</b>            | <ul style="list-style-type: none"> <li>• System is online</li> <li>• Subscriber is authenticated and logged in</li> <li>• Subscriber has an active subscription</li> </ul> |
| <b>Postconditions</b>           | <ul style="list-style-type: none"> <li>• Recurring charges are stopped and the account is scheduled to downgrade at the end of the billing cycle</li> </ul>                |
| <b>Actors</b>                   | <ul style="list-style-type: none"> <li>• Subscriber</li> <li>• Payment Gateway</li> </ul>                                                                                  |
| <b>BASIC FLOW</b>               | <ol style="list-style-type: none"> <li>1. Subscriber navigates to Billing Settings.</li> <li>2. Subscriber selects “Cancel Subscription.”</li> </ol>                       |

| USE CASE ELEMENT                                                        | SPECIFICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         | <ol style="list-style-type: none"> <li>3. System prompts for confirmation and an optional cancellation reason.</li> <li>4. Subscriber confirms.</li> <li>5. System notifies the Payment Gateway to stop future recurring charges.</li> <li>6. System schedules the downgrade to Registered Reader tier at the end of the current billing cycle.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>ALTERNATE FLOW<br/>A1 (Rescind<br/>Cancellation)</b>                 | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                                                                                                                                                                                                                              |
| <b>EXCEPTION FLOW<br/>E1 (Gateway<br/>Acknowledgment<br/>Failure)</b>   | <ol style="list-style-type: none"> <li>1. The user reaches the condition described by this flow.</li> <li>2. The system detects the condition during the relevant basic-flow step.</li> <li>3. The system validates the current record, permission, and business rules.</li> <li>4. The system performs the permitted alternate handling or blocks the invalid operation.</li> <li>5. The system explains the resulting state and required next action.</li> <li>6. The system records the event when audit/security rules require it.</li> <li>7. The user continues from the appropriate point after the condition is resolved.</li> </ol>                                                                                                                                                                                                                                                                                                                              |
| <b>EXCEPTION FLOW<br/>E-AUTH (Authorization<br/>or Session Failure)</b> | <ol style="list-style-type: none"> <li>1. The user initiates the protected operation.</li> <li>2. The system validates the current session and required role or ownership permission immediately before the state-changing action.</li> <li>3. The system detects an expired session or missing permission.</li> <li>4. The system blocks the state change and does not partially commit it.</li> <li>5. The system records the denied attempt when required by security policy.</li> <li>6. The system redirects to authentication or displays access denied.</li> <li>7. After authorization is restored, the user may retry.</li> </ol> <p>Revision Note: Expanded every documented Alternate and Exception Flow across all nine modules into explicit numbered procedures. UC-1.4 and UC-1.8 now support Editor-requested revisions for both Writer articles and Graphics Designer magazine designs. Module 9 contains no promo-code or discount-code processing.</p> |